

STEEL FAB ENTERPRISES

SFE Milestone Tracking & Estimation System

Complete A-to-Z Application Workflow & Functionality Reference

Master Knowledge Base for the In-App AI Assistant ("Steel Fab Assist"), QA Reference, and User Training Manual

Document Type: Functional & Navigational Reference Manual

Scope: Every module, screen, tab, button, field, modal, validation rule, and workflow

Audience: End users, support engineers, QA, developers, and the RAG chatbot itself

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PART 1 — INTRODUCTION & SYSTEM OVERVIEW

1.1 Purpose of This Document

This manual is the single authoritative reference describing exactly how the SFE Milestone Tracking & Estimation System behaves from the perspective of a real user clicking through the application. It documents every module, every screen inside every module, every visible button, tab, dropdown, modal, table, filter, and form field, and explains what happens when a user interacts with each of them. It is written specifically to be ingested as the knowledge base for the application's built-in AI assistant ("Steel Fab Assist"), so that the assistant can answer module-level, screen-level, and button-level questions with precision. The same document equally serves as an onboarding guide for new users, a troubleshooting reference for support engineers, a QA test-reference, and a functional specification for developers and product owners.

Each section below is written to be reasonably self-contained: a screen description restates which module it belongs to and how a user reaches it, so the information remains understandable even if only a fragment of this document is retrieved in isolation by the assistant.

1.2 What the Application Is

The SFE Milestone Tracking & Estimation System is an internal enterprise web application built for Steel Fab Enterprises (SFE), a structural steel fabrication and erection contractor. It digitizes the complete lifecycle of a structural steel project: capturing a Request for Quotation (RFQ) from a prospective client, tracking the internal bid-preparation calendar, producing a detailed cost and weight estimate, recording whether the bid is won or lost, converting a won bid into an active Project, scheduling that project's structural detailing and shop-drawing sequences, releasing fabrication work to the shop floor in priority order, and mapping the shop's machine and labor capacity against the production demand created by all active projects. A single web portal exposes all of this as one connected system rather than a set of disconnected spreadsheets.

1.3 Technology Architecture

- **Frontend:** A React single-page application built with Vite, styled with Tailwind CSS, using Lucide icons throughout, React Router for navigation (every route is mounted under the base path `"/SFE"`), TanStack React Query for data fetching/caching on the newer RFQ screens, and AG Grid for the large RFQ spreadsheet.
- **Backend:** A Django REST Framework API serving JSON to the frontend, organized into focused Django apps (accounts, projects, employees, bids, production, milestones, dashboard, chatbot) plus a family of sub-modules under `"apps/"` for the Estimation & Erection tabs (core, breakdown, erection_takeoff, field_moment_conn, misc_metals, estimate_data, rfq, bid_summary, dashboards).
- **Database:** SQLite for local/development use and PostgreSQL in production deployments.
- **Authentication:** Session/token-based login; the authenticated user object — including role and the list of permitted module keys — is cached in the browser's `sessionStorage` and consulted by every protected route and by the sidebar to decide what to display.
- **AI Assistant:** A Retrieval-Augmented Generation (RAG) chatbot. Administrator-uploaded PDF documents (such as this one) are parsed page by page and split into roughly 800-character overlapping text chunks, which are indexed with BM25 keyword ranking. When a user asks a question, the most relevant chunks are retrieved and passed to a language model (an offline Ollama `"llama3.2"` model, or a configured cloud model) to produce a grounded answer, with citations to source page numbers. The assistant can also call a small set of tools to take real actions inside the app — see Part 14.

1.4 High-Level End-to-End Business Workflow

A user can jump directly to any module they have access to, but the natural order in which work flows through the system is as follows. Each stage is documented in full in its own Part of this manual.

1. A prospective job arrives and is logged as a Request for Quotation (RFQ) in the RFQ Master / Data Entry grid (Part 5), with a unique Quote Number, customer, scope of work, and bid due date.
2. The internal team plans its review and preparation milestones for that bid on the Internal Bid Schedule calendar (Part 5.9), which itself respects non-working days defined in the Holiday Calendar (Part 5.10).
3. An estimator opens the RFQ inside the Estimation & Erection module (Part 6) and works through the Contracts, Estimation Model, Erection Take-off, Field Moment Connections, Misc Metals, Breakdown, and Estimate Data tabs to build a complete, line-item cost and weight estimate culminating in a single Final Bid Amount.
4. Once accepted by the client, the estimator marks the RFQ "Won" in the Data Entry grid; separately, the Estimate Data tab lets the team finalize/lock the estimate with a digital signature, producing an immutable revision snapshot.
5. The Won RFQ is used to pre-fill a new entry in Project Master (Part 9), which becomes the live system-of-record project. Project Master also captures the project's detailed Structural Schedule — a sequence-by-sequence grid of tonnage, drawing-approval dates, RTS (Ready-to-Ship) dates, and fabrication/field man-hour budgets.
6. The same structural sequence data is managed and visualized for sequencing purposes in the Structural Schedule module (Plan Creation and Plan Tracking, Part 10), including a Gantt-chart view of every sequence's timeline and progress.
7. As sequences approach fabrication, they are released to the shop floor through Production Management (Part 11): the Production Priority Schedule assigns job/sequence work to machine routing stations (Plate, Angle, and Structural priority queues) with start/complete run dates.
8. Capacity Mapping (Part 12) cross-checks the shop's available machine and manpower capacity, by shop/location and by month, against the demand generated by all active production schedules, surfacing over- or under-utilization.
9. Throughout, Dashboard (Part 4) gives both an operations view (today's sequence health, milestones, capacity, and tonnage) and an executive/VP view (bid win-rate, pipeline value, and forward capacity outlook) of the same underlying data.

1.5 User Roles & Permissions

Access control in the application is enforced on two independent layers, and a real user is governed by both simultaneously.

1.5.1 The Six User Roles

Every user account carries exactly one role, chosen at account-creation time on the User Access screen (Part 13.3). The roles are:

Role (stored value)	Display Name	Typical Function
admin	Admin	System administrator. Full, unrestricted access to every screen regardless of the module-permission list, plus exclusive access to the User Access screen and the chatbot's document-upload Context Manager.

Role (stored value)	Display Name	Typical Function
manager	Manager	Operations manager. Sees both the Executive and Operations dashboards (with a toggle switch) and is treated as a manager/editor throughout the bid and production screens.
estimator	Estimator	Bid estimator. Primary user of RFQ logging, the Estimation & Erection module, bid calendars, and bid results; sees the Executive dashboard view.
detailing	Detailing / JF	Detailing partner role focused on structural sequencing and detailing-related screens; sees the Executive dashboard view.
employee	Employee	Shop-floor / operational staff; sees the Operations dashboard view (sequence health, schedules) rather than the bid/executive view.
readonly	Read-Only	Audit/guest role with view access; sees both dashboard views but cannot create, edit, or delete records.

1.5.2 Per-Module Access Control

Independently of role, every non-admin user has an "allowed_modules" list — an explicit array of module keys (e.g. "/dashboard", "/rfq", "employees", "/production/priority-schedule") stored on their account and editable only by an Admin from the User Access screen. A brand-new account is seeded with a default list covering nearly all modules except User Access. The Sidebar reads this list and hides any navigation item whose module key is not present; the routing layer (the ProtectedRoute wrapper) performs the same check a second time when a route is opened directly by URL, redirecting an unauthorized user to the first module they are allowed to see, or to the Dashboard. Administrators bypass this check entirely and always see every module. The "User Access" sidebar entry itself is only ever shown to Admins, regardless of any module list (it is an admin-only route).

In short: a user must (a) hold a role that is not blocked from a given page's logic, and (b) have that page's module key present in their allowed_modules list, in order to see and use that page. The complete module-key list is reproduced in Appendix B.

1.5.3 Dashboard Visibility by Role

The Dashboard module (Part 4) is special in that it has two different "modes" — an Executive/Bid-and-Estimation view and an Operations view — and which ones a given user can see (and which one loads by default) depends directly on role:

Role	Can see Executive (Bid & Estimation) view?	Can see Operations view?	Default view shown
admin	Yes	Yes	Executive
manager	Yes	Yes	Operations
estimator	Yes	No	Executive
detailing	Yes	No	Executive
employee	No	Yes	Operations
readonly	Yes	Yes	Operations

When a user can see both views, a pill-shaped switcher control appears at the top of the Dashboard letting them toggle between "Bid and Estimation" and "Operations" at will (see Part 4.1).

1.6 Application-Wide UI Conventions

The following interaction patterns recur across almost every module in the system, so they are explained once here rather than repeated on every screen.

1.6.1 Feedback Messages: Toasts, Inline Banners, and Browser Dialogs

- **Toast notifications:** Small messages that slide in from the top-right corner of the screen and automatically disappear after about 3 seconds. Newer screens (RFQ Data Entry, Holiday Calendar, Process Master Settings, User Access, Estimation Model, Dollar Dashboard, Estimation Summary) use green toasts for success messages ("Saved", "Holiday updated", "User registered successfully") and red toasts for errors ("Failed to load settings data.", "Delete failed").
- **Inline form banners:** A red or green banner embedded directly inside a form (for example on Login, Forgot Password, Reset Password, Announcements, and Settings) that stays visible until the user takes another action.
- **Native browser confirm dialogs:** Older/simpler screens (Employee Master, Customer Master, Detailer Master, Project Master, Plan Creation, Capacity Mapping, Production Priority Schedule, Process Master, Chatbot session deletion) pop up the browser's native confirmation dialog before any destructive delete action — for example "Are you sure you want to delete this employee?" — and a native alert popup to report a failure such as "Failed to delete employee" or "Failed to save customer".

1.6.2 Modals

Almost all "Add" and "Edit" actions throughout the system open a centered modal dialog over a darkened, blurred backdrop, with a header (title plus a close "X" control), a scrollable body containing the form fields, and a footer with a secondary "Cancel" button and a primary action button (commonly styled in the application's signature amber-to-orange gradient). Clicking outside the modal does not close it — only the "X" or "Cancel" controls do, which is deliberate, to avoid accidental loss of entered data on long forms.

1.6.3 Searchable Dropdowns

Most "select a record" fields (selecting a customer, detailer, employee, project, machine, etc.) use a custom searchable-dropdown control rather than a plain HTML select box: clicking the field opens a panel with a text box at the top that filters the option list as the user types, plus, depending on context, an "X" control to clear the current selection.

1.6.4 Formatted Date Inputs

Date fields across the system (schedules, holidays, RFQ dates, etc.) use a formatted date-input control that displays and accepts dates in MM-DD-YYYY format while storing them internally as ISO dates (YYYY-MM-DD); the control auto-inserts the dashes as the user types digits.

1.6.5 Auto-Save Indicator

The Estimation & Erection tabs (Contracts, Estimation Model, Erection Take-off, Field Moment Connections, Misc Metals, Breakdown, and Estimate Data) display a small auto-save badge near the top-right of the screen showing a live state — idle, saving (with a spinner), saved (with a checkmark and the last-saved time), or error — because field edits on those tabs are silently persisted to the server shortly after the user stops typing, without an explicit Save click for most individual fields.

1.6.6 Status Badges & Color Language

Status pills appear throughout the system as small rounded badges, with a broadly consistent color vocabulary: green/emerald for "Completed," "Won," "Active," or "On Track"; blue for "In Progress" or "Upcoming"; amber/orange/yellow for "Likely Delay," "Due Today," "Pending," or items awaiting attention; red/rose for "Delayed," "Lost," "Overdue," or destructive actions; and grey/slate for "Yet to Start," "Inactive," "No Date," or disabled states.

1.6.7 Tables: Search, Filter, Sort, Paginate, Export

List/table screens generally provide a free-text search box (filtering client-side on the records already loaded, typically matching name/code/customer fields), one or more dropdown filters (status, department, role, etc.), and on longer lists, numbered pagination controls (commonly 10 records per page, with First/Previous/Next/Last icon buttons). A Download/Export control, where present, exports the currently visible (filtered) rows to a CSV file, or in a few places (Project Master, Plan Creation, the structural schedule grid) to a formatted PDF.

1.6.8 Disabled / Read-Only States

Fields are disabled (greyed out, not editable) in three recurring situations: (1) a "View" modal opened in read-only mode; (2) data that is synced or derived from another module and therefore not directly editable here (for example the Customer Name field on the Project form, which is "Synced from RFQ"); and (3) any Estimation & Erection tab belonging to a project whose estimate has been finalized/locked, at which point every tab in that module becomes read-only system-wide for that project.

1.7 How to Read This Document

Each subsequent Part covers one functional module of the application. Within a Part, each screen is described with: its purpose, how a user navigates to it, who can access it, every visible element (cards, tables, columns, buttons, tabs, filters), and an explicit "user clicks X → system does Y" mapping for every interactive control, including the exact validation and success/error messages the system shows, where these were verifiable in the underlying application code.

PART 2 — GETTING STARTED — AUTHENTICATION

Before reaching any part of the application, every user must pass through the public Authentication module, which consists of three screens: Login, Forgot Password, and Reset Password. These screens are reachable without being signed in, and the application automatically redirects an already-logged-in user away from them back to the Dashboard.

2.1 Login Screen

Navigation Path: Default landing page at `"/login"` for any unauthenticated visitor; also reached automatically whenever a session expires or after Sign Out.

Purpose: Authenticates a user against their email and password and establishes their session.

Layout: A two-pane screen on desktop: the left pane is a dark branding panel with a rotating photo slideshow (four images, auto-advancing every 5 seconds, with clickable dot indicators) and the "Steel Fab ENTERPRISES — Milestone Management System" wordmark; the right pane holds the actual sign-in card. On narrower/mobile screens, only the sign-in card is shown, with a compact mobile-only logo and wordmark above the form.

The sign-in card contains the following elements:

- **Email Address field:** a text input, placeholder `"you@steelfab.com"`.
- **Password field:** a masked text input with a show/hide "eye" icon toggle that switches the field between hidden dots and plain text.
- **Remember me checkbox:** a checkbox to the left of the form footer (present in the UI; does not change the underlying session duration).
- **Forgot password? link:** routes to the Forgot Password screen.
- **Sign In button:** a gradient amber-to-orange submit button. While the request is in flight it shows a spinning loader and the label "Signing in..."; otherwise it reads "Sign In" with a right-pointing arrow icon.

Action: Login Submission

- **User Action:** the user fills in Email Address and Password, then clicks "Sign In" (or presses Enter).
- **Validation:** if either field is empty, the form blocks submission and shows the inline message "Please fill all required fields."
- **System Result on success:** the application calls the login API; the returned user record (including role and `allowed_modules`) is stored in the browser's `sessionStorage`, and the user is redirected to the Dashboard (`"/dashboard"`).
- **Error Scenario — network unreachable:** "Cannot reach server. Please check your internet connection."
- **Error Scenario — server error (HTTP 500):** "Server Error (500): Please ensure database migrations have been run on the backend."
- **Error Scenario — invalid credentials or other validation failure:** the system surfaces the most specific message it can find in the server response (for example an email- or password-specific error), falling back to "Failed to process request. Please check your details."

All error and validation messages render as a red banner directly beneath the password field, above the Sign In button.

2.2 Forgot Password Screen

Navigation Path: Login screen → "Forgot password?" link → `"/forgot-password"`.

Purpose: Initiates the password-reset process by emailing the user a one-time code (OTP).

Layout: a centered white card showing the Steel Fab logo/wordmark, the heading "Forgot password?", the helper text "Enter your email and we'll send you a reset link.", an Email Address input (placeholder "you@steelfab.com", required), a "Send Reset Link" submit button, and a "Back to sign in" link beneath the form.

Action: Request Reset Code

- **User Action:** the user enters their email address and clicks "Send Reset Link".
- **System Result:** on success, the form switches to a confirmation state — a green checkmark icon, the heading "Check your email", and the text "We've sent a password reset link to <email>." — and after about 1.5 seconds the browser automatically redirects to the Reset Password screen with the email pre-filled in the URL.
- **Error Scenario:** if the request fails, a red banner shows the server-provided message, or the fallback "Failed to send reset link."

2.3 Reset Password Screen

Navigation Path: Reached automatically from Forgot Password, or directly via an emailed link of the form "/reset-password?email=...".

Purpose: Validates the emailed One-Time Password (OTP) code and sets a new account password.

Layout: a centered card with the heading "Reset password" and the fields:

- **Email Address:** pre-filled from the URL if the user arrived via the Forgot Password flow, but still editable.
- **OTP Code:** a 6-digit, centered, monospace text field, placeholder "000000", limited to 6 characters.
- **New Password:** a masked field with a show/hide toggle, placeholder "Min 6 characters".
- **Confirm Password:** a masked field, placeholder "Re-enter password".
- **Reset Password button:** submits the form; shows a spinner and "Resetting..." while in flight.

Action: Save New Password

- **Validation — missing email:** "Email is required."
- **Validation — password mismatch:** if New Password and Confirm Password do not match, the system shows "Passwords do not match" and does not submit.
- **Validation — password too short:** if the new password is fewer than 6 characters, the system shows "Password must be at least 6 characters" and does not submit.
- **System Result on success:** the form switches to a success state — a green checkmark, the heading "Password reset!", the text "You can now sign in with your new password.", and a "Back to sign in" link back to Login.
- **Error Scenario:** the server's error message is shown (for example for an invalid or expired OTP), falling back to "Reset failed. Please try again."

2.4 Session & Sign-Out Behavior

Once logged in, the user's session is represented purely by the user object cached in sessionStorage (closing the browser tab or clearing storage effectively ends the session). "Sign Out" is available from two places — the Sidebar's user-profile footer (a logout icon) and the Header's profile dropdown ("Sign Out" with a logout icon) — and both perform the same action: call the logout API, clear all of sessionStorage, and redirect the browser to the Login screen.

2.5 Protected Routes & Module-Based Redirects

Every screen other than Login, Forgot Password, and Reset Password is wrapped by a route guard. If no user is found in sessionStorage, the guard immediately redirects to Login. If a route declares that it requires a specific module key (almost every screen does) and the signed-in user is not an Admin and does not have that module key in their allowed_modules list, the guard silently redirects the user to the first module on their allowed list (or to the Dashboard if their list is empty) rather than showing an access-denied page. A small number of routes (currently only User Access) are marked admin-only and redirect any non-admin in the same way regardless of their module list.

PART 3 — GLOBAL APPLICATION SHELL

Every protected screen in the application is rendered inside a common shell consisting of a left Sidebar, a top Header bar, and a floating Chatbot widget in the bottom-right corner. This Part documents that shell once; it is present, with only minor labeling differences, on every single screen described in the rest of this manual.

3.1 Sidebar Navigation

The Sidebar is a fixed panel on the left edge of the screen, with a dark navy gradient background. It contains, top to bottom: the SFE logo and "Fabrication Project Management" wordmark; a scrollable navigation list grouped into nine labeled sections; a user-profile footer with avatar, name, email, and a Sign Out icon button; and, on desktop only, a small circular amber collapse/expand toggle button on the sidebar's right edge.

3.1.1 Behavior

- **Desktop collapse:** clicking the amber chevron toggle shrinks the sidebar to a narrow 78-pixel icon-only rail (hiding section labels and item text, but showing a tooltip with the item name on hover) or restores it to its full 240-pixel width.
- **Mobile drawer:** on narrow screens the sidebar is hidden off-canvas by default and slides in as an overlay (with a dimmed, blurred backdrop) when the Header's "Menu" icon is tapped; tapping the backdrop, the in-sidebar "X" close icon, or selecting any navigation item closes the drawer again.
- **Active item highlighting:** the navigation item matching the current page is highlighted and its icon turns amber.
- **Role-based filtering:** Admin users see every section and every item. All other users see only the items whose module key is present in their personal allowed_modules list; a section is hidden entirely if none of its items are visible to the current user. The "User Access" item is shown only to Admins.

3.1.2 The Nine Navigation Sections

The Sidebar groups its links into the following sections, in this fixed order:

Section	Items (in display order)
OVERVIEW	Dashboard
BID MANAGEMENT	RFQ Master · Internal Bid Schedule · Holiday Calendar · Estimation Summary · Estimation
MANAGEMENT	Employee Master · Customer Master · Detailer Master
BUDGET ESTIMATOR	Design Inputs · Estimation Result
PROJECT MANAGEMENT	Project Master
STRUCTURAL SCHEDULE	Plan Creation · Plan Tracking
PRODUCTION MANAGEMENT	Production Schedule · Process Master Settings
CAPACITY MAPPING	Capacity Configuration · Machine Master · Workforce Master
SYSTEM	Settings · Announcement · User Access (Admins only)

Note that "RFQ Master" is the entry point to the entire six-tab RFQ sub-application (Part 5.1), and "Estimation" is the entry point to the seven-tab Estimation & Erection module (Part 6). "Production Schedule" is itself a two-tab screen (Production Priority Schedule and Process Master, Part 11).

3.2 Header Bar

A sticky bar fixed to the top of the content area on every screen, divided into three equal zones.

- **Left zone:** a "Menu" (hamburger) icon, visible only on mobile, that opens the Sidebar drawer; and a "Back" button (a left-pointing arrow, with the word "Back" on wider screens) that navigates the browser one step back in history.
- **Center zone:** the current page's title, computed automatically from the active route (for example "Dashboard," "Employee Master," "RFQ Data Entry," "Production Schedule," or "Estimation" while anywhere inside the Estimation & Erection module).
- **Right zone:** a live clock showing the current time (HH:MM:SS) and full date, visible on extra-large screens; a notification Bell icon that opens a dropdown panel (currently always showing "No new notifications — You're all caught up!" since no notifications are generated by the system yet); and a circular profile avatar (showing the user's photo if one is set, otherwise their initials) with a chevron, which opens a small dropdown showing the user's name and a "Sign Out" option.

3.3 Floating AI Chatbot Widget

A circular orange "Steel Fab Assist" launcher button is fixed to the bottom-right corner of the viewport on every screen. A small pulsing green dot appears on the button whenever the backend AI/Ollama server is reachable. Clicking the button toggles a 400-by-600-pixel chat panel open above it; the same button (now showing an "X") closes the panel.

A full description of the chatbot's capabilities, including the document-upload Context Manager and its agentic tools, is given in Part 14. In summary, the panel provides:

- **A session sidebar:** a vertical 80-pixel-wide strip on the left of the panel listing the user's chat threads as small tiles (title + message icon), a "+ New" button to start a fresh thread, and a per-tile trash icon (with a confirm prompt, "Delete this chat thread?") to remove a thread.
- **A message area:** shows the conversation as left-aligned assistant bubbles and right-aligned user bubbles; when no messages exist yet it instead shows a welcome panel with three clickable suggested prompts ("How does the RFQ workflow operate?", "How do I create a new project?", "What are the steps in Erection Estimation?").
- **A text input bar:** a single-line text box (placeholder "Ask a question..." or, if the assistant server is unreachable, "Waiting for Ollama...") and a Send button; the input is disabled while the assistant server is offline.

PART 4 — DASHBOARD MODULE

Navigation Path: Sidebar → OVERVIEW → Dashboard ("/dashboard"). This is also the page a user lands on immediately after logging in.

The Dashboard is the system's home screen and is built as two largely independent sub-views — an "Executive" (labeled "Bid and Estimation") view and an "Operations" view — sharing the same page. Which view(s) a given user can see, and which one loads by default, is governed strictly by role (see Part 1.5.3).

4.1 The Mode Switcher

If — and only if — the signed-in user is permitted to see both views (this applies to the admin, manager, and readonly roles), a pill-shaped switcher control appears at the top of the page with two tabs: "Bid and Estimation" (a crown icon) and "Operations" (a dashboard icon). Clicking either tab instantly swaps the entire page content to that mode; the inactive mode's data is simply hidden, not unmounted, so switching back and forth does not require reloading data. Users who can see only one of the two views never see this switcher; their single available view simply renders directly.

4.2 Operations Dashboard

The Operations view is built around the live state of every structural sequence across every project (the same Structural Schedule data described in Part 9 and Part 10), reclassified here for at-a-glance health monitoring.

4.2.1 The Four Stat Cards

Four large clickable cards sit at the top of the Operations view. Every structural sequence in the system is automatically placed into exactly one of three pipeline stages based on which dates have been recorded on it: "Under Detailing" (the sequence has a planned RTS date but no actual RTS date yet), "Fabrication" (the sequence has an actual RTS date but no actual ship date yet — shown with a sub-status of "Released for Fabrication" or, once a ship date target exists, "Under Fabrication"), and "Sent for Erection" (the sequence has an actual ship date recorded).

Card	Shown Value	Caption
Total Projects	Total number of projects in the system	Total sequence count across all projects
Under Detailing	Count of sequences awaiting actual RTS	Number of distinct projects with at least one such sequence
Fabrication	Count of sequences released to or in fabrication	Number of distinct projects with at least one such sequence
Sent for Erection	Count of sequences shipped	Number of distinct projects with at least one such sequence

Action: Drill Into a Stage

- **User Action:** the user clicks any one of the four cards.
- **System Result:** the clicked card becomes visually "Selected" (highlighted border and accent strip) and a details table appears beneath the cards, titled "All Active Sequences," "Under Detailing Sequences," "Fabrication Sequences (Clubbed)," or "Sent for Erection Sequences" to match. The table lists every matching sequence with its project, sequence number, description, and computed sub-status, and includes a search box (placeholder "Search by project or sequence...") that filters within the selected stage.

- Clicking the same card a second time deselects it and collapses the table.

4.2.2 Project Health Hierarchy

A panel titled "Project Health Hierarchy" ("Status grouped by schedule adherence · Click to expand") classifies every sequence into one of six schedule-adherence states and presents them as a row of clickable filter pills, each showing a live count: Yet to Start, On Track, Completed, Delayed, and Likely Delay (a sequence is "Likely Delay" if its planned RTS date has passed within the last 7 days without an actual RTS/completion recorded, and "Delayed" once that gap exceeds 7 days; "Completed with Delay" is used for a sequence that finished later than its original plan). Clicking a pill filters the panel below to that status; the panel then lists every project containing at least one sequence in that status, and clicking a project row expands it into a detailed per-sequence table with columns Seq No, Description, RTS Date, Actual RTS, Status (color badge), and Tons.

4.2.3 Milestone Management (Gantt Chart Widget)

A wide panel titled "Milestone Management" with the sub-label "Gantt Chart • Project Timelines & Schedules" renders an interactive Gantt chart of every project's structural-schedule sequences across a selectable time window. Controls in the panel header let the user choose a "From" month, a "To" month (constrained so it cannot be set earlier than "From"), and a year (selectable from 2012 up to four years in the future), and the chart redraws to that window. This Gantt widget is the dashboard's real, functional view of project milestones; it should not be confused with a separate, currently unused "Milestone Management" page described in Part 16.3.

4.2.4 Capacity Utilization Summary

A card titled "Capacity Utilization Summary" ("Manhours loading") provides two view modes — "Monthly Planning" and "Yearly Forecast," selected with an inline tab toggle — plus month and year dropdown selectors (the month selector is disabled while in Yearly Forecast mode). For the selected period it shows four KPI tiles: Available Hours (the shop's workforce capacity for that period), Required Hours (the active man-hour load created by scheduled work), a Surplus or Shortage Hours tile (colored green when capacity exceeds demand, rose/red when demand exceeds capacity), and a capacity-utilization percentage gauge.

4.2.5 Tonnage Loading Summary

A card titled "Tonnage Loading Summary (<year>)" shows monthly tonnage loading for the selected year (with its own year dropdown, also ranging 2012 to four years ahead) as a chart, alongside the total allocated tonnage versus total won tonnage for context.

4.2.6 Operations Log

A panel titled "Operations Log" lists a chronological feed of recent system activity entries, each shown with an icon colored by type (green check for success, amber triangle for warning, grey clock for informational), an action description in capital letters, and the associated project name and a relative time stamp.

4.2.7 Internal Notices

A panel titled "Internal Notices" lists currently active system Announcements (see Part 13.2), each as a single line with a priority dot (red = high, amber = medium, grey = low/normal) and the announcement title.

Action: View a Notice

- User Action: the user clicks any notice line in the Internal Notices panel.
- System Result: a modal opens showing the notice's full title, its full message body, and — if set — its active date range ("Active: <from> to <to>"). A close "X" button or clicking the dark backdrop dismisses the modal.

4.3 Executive / VP Dashboard ("Bid and Estimation")

The Executive view reframes the same underlying project and bid data for sales/leadership use, centered on bid pipeline health, win performance, and forward capacity rather than shop-floor sequencing.

- **Activity calendar:** an interactive calendar plotting upcoming and recent dates of three types — Bid Due Date, Quote Submitted Date, and Awarded Date — each rendered with its own colored dot/badge, so the team can see at a glance which bids are coming due, which quotes were recently submitted, and which jobs were recently awarded.
- **Executive Advisor:** a capacity-outlook panel (sub-labeled with the currently selected month and year) comparing Available man-hours against Required man-hours to give a forward-looking staffing recommendation.
- **Bid Performance Analytics:** a panel titled "Bid Performance Analytics" ("Total Quoted vs Won") rendering a chart of total quoted bid value against total won bid value over time, with a toggle to view the trend year-over-year, alongside simple counters for Total, Won, Lost, and Pending bids.
- **Latest Quotes and Latest Wins:** two running lists surfacing the most recently logged RFQ quotes and the most recently won bids, for quick visibility into fresh pipeline activity.
- **Load Utilisation:** a gauge-style indicator summarizing overall shop load utilization, mirroring the same available-vs-required-hours concept used in the Operations view's Capacity Utilization Summary but framed for an executive audience.

PART 5 — BID MANAGEMENT MODULE (RFQ)

The Bid Management module is where every prospective job enters the system as a Request for Quotation (RFQ), is tracked through internal review, costed, and ultimately marked Won or Lost. It comprises the RFQ Master screen (itself a six-tab sub-application plus a separate Print Setup screen), the Internal Bid Schedule calendar, and the Holiday Calendar. The related Estimation Summary screen, which lists every estimate ever produced for an RFQ/project and supports the "rebid" workflow, is documented at the end of this Part for convenience since it is reached from the same sidebar group.

5.1 RFQ Master

Navigation Path: Sidebar → BID MANAGEMENT → RFQ Master ("/rfq"), which defaults to its Data Entry tab.

RFQ Master is not a single screen but a layout shell with a strip of six tabs running along the top: Data Entry, Bid Performance, Dollar Dashboard, Job Analytics, Sales Cycle, and Future Capacity. All six tabs share the same set of underlying RFQ/bid records; the Data Entry tab is where records are created and edited, while the remaining five are read-only analytical views computed from that same data.

5.2 Data Entry Tab — The RFQ Spreadsheet

Navigation Path: RFQ Master → Data Entry tab ("/rfq/data-entry").

Purpose: The master log of every RFQ/bid the company has quoted: an enormous, fully editable spreadsheet (built with AG Grid) with roughly 80 columns, organized into thirteen logical column groups, covering everything from initial bid intake through award and fabrication start.

Toolbar

- **Record/status summary:** shows the total record count plus a live breakdown of how many are Won, Lost, and "In Progress" (the system's internal "Pending" status is displayed to the user as "Pending"; note the header summary text itself uses the word "Progress").
- **" + New RFQ " button:** visible only to users with edit rights; opens the New RFQ modal (described below).
- **"Send Emails" button:** visible to editors; shows a numeric badge counting RFQs that have a Scope of Work set but have not yet had their project-details email sent. Disabled when that count is zero. Clicking it opens a "Confirm Bulk Send Emails" modal; confirming triggers the bulk-email API and shows the toast "Bulk emails triggered successfully" (or the server's specific error message on failure).
- **"Upload Excel" button:** visible to managers/editors; opens the Excel Upload modal (described below). Tooltip: "Upload Excel file to import records."
- **"Export" button:** exports the grid's current rows to a CSV file named "SFE_RFQ_Export_<date>.csv".
- **Refresh icon:** re-fetches the grid data from the server.
- **Search box:** placeholder "Search quote, project, customer...", with "Search" and "Clear" buttons; pressing Enter also triggers the search. Applying a search updates the URL's query parameters so the filtered view is shareable/bookmarkable.

The Spreadsheet Column Groups

Every row is one RFQ. Columns are grouped under the following headers (left to right); most cells are directly editable in place by clicking into them, and an edit is saved automatically with a "Saved" toast, or a red toast listing the field-level validation errors returned by the server if the save fails.

Column Group	Representative Columns
Project Identity	Row Actions (see below) · Quote No · Bid Reference · Project Name
Bid Details	Type · Project Comments · Bid Due (date) · Time · Location · Distance · Age (days since the record was created, colored amber past 30 days and red past 90) · Scope of Work · Rebid?
Project Flags	AISC Fab · AISC Erect · Domestic Steel · LEED · Minority Participation · Prevailing Wage · CCIP/OCIP · Bonded · Paint · Galvanized · Professional Engineer (PE) · 3rd Party Inspection · Tax Status
Customer & Team	Customer · Decision (to bid) · Estimator (initials) · Outsourced Estimator · sent-to-J&D / sent-to-Detailing / sent-to-Erection flags · Est. Sqft/Ton
Pricing	Structure \$ · Erection \$ · Misc \$ · Misc Erection \$ · Bid Amount · Quoted Profit
Quantities	Ton Steel · Ton Joist · Total Tons (computed) · # Structural Pieces · Sq Ft
Unit Costs (Computed, read-only)	Pieces/Ton · Tons/Sqft (no joist) · Tons/Sqft (with joist) · \$/Ton Fab · \$/Ton Erect · \$/Sqft Fab · \$/Sqft Erect
Structural Fab Schedule	Hours · Start Month · Duration (months) · End Month (computed) · Avg Hours/Month (computed)
Misc Fab Schedule	Hours · Start Month · Duration · End Month (computed) · Avg Hours/Month (computed)
Structural Erection Schedule	Hours · Start Month · Duration · End Month (computed) · Avg Hours/Month (computed)
Misc Erection Schedule	Hours · Start Month · Duration · End Month (computed) · Avg Hours/Month (computed)
Estimating & Outcome	Estimating Hours · Quote Date · Won/Lost · Follow-up Date · Follow-up Notes · Awarded Amount
Post-Award	SFE Job # · Award Date · Contract Date · Fabrication Start Date

The "End Month" and "Avg Hours/Month" columns in each of the four schedule groups are read-only because the server automatically derives them from that group's Start Month plus Duration, and from its Hours divided by Duration, respectively, the moment the row is saved (this calculation logic mirrors the same auto-calculation performed inside the Estimation Model worksheet — see Part 6.3).

Row Actions

A compact set of icon buttons appears in the pinned left-most column of every row:

- **Duplicate as Rebid (copy icon):** clones the entire RFQ record into a brand-new row (a fresh Quote Number is generated) so the team can re-quote a lost or stale bid without retyping its specifications. Shows the toast "Rebid created: <new quote number>" on success.
- **Send / Resend Email (mail icon):** sends the project-details email for that single RFQ. Disabled (with the tooltip "Please select a Scope of Work first") until the row has a Scope of Work set. Once an email has been sent, the icon switches to a green "sent" styling with the tooltip "Resend Project Details Email." Clicking it opens a "Confirm Send Email" modal; confirming shows "Email triggered successfully."
- **Set Job # (briefcase icon, labeled "Job #"):** appears only on rows that are Won but do not yet have an SFE Job Number assigned. Opens the small "Set SFE Job #" modal (a single numeric field, e.g. "1042") whose Save records the job number and shows the toast "SFE Job # set."

- **Awarded indicator:** once a Won row has a Job # assigned, the action area instead shows a static green "✓ #<job number>" badge in place of the Set Job # button.
- **Delete (trash icon):** prompts the native confirmation "Do you really want to delete?"; on confirm, deletes the record and shows "Record deleted," or "Delete failed" on error.

Action: Create a New RFQ

- **User Action:** click "+ New RFQ".
- **System Result:** the "New RFQ" modal opens with fields for Quote No (placeholder format e.g. "26-05-01", required), Type, Reference #, Distance to Travel, Customer (searchable dropdown, "— Select Customer —"), Decision (to bid), Scope of Work, and Primary Estimator (searchable dropdown, "— Select Estimator —").
- **On Save,** the system creates the record, shows the toast "RFQ created," closes the modal, and the new row appears in the Data Entry grid for further (now spreadsheet-style) editing of all its remaining fields.
- **Error Scenario:** a red toast shows the server-reported validation message.

Action: Bulk Import via Excel

- **User Action:** click "Upload Excel", choose a local .xlsx file in the file picker inside the "Upload Excel" modal.
- **Validation:** if the chosen file is not a .xlsx file, the system rejects it immediately with "Only .xlsx files are accepted."
- **System Result:** the file is uploaded for a preview pass; on success the modal reports how many records are new versus how many will update existing ones, e.g. "Import complete: 12 new, 4 to update," and the matched RFQs are merged into the grid.
- **Error Scenario:** a toast shows the server's specific import error.

Action: Pull from SEBW

A "Pull from SEBW" modal (reachable from within the RFQ workflow) lets a user manually enter Steel Estimate / Bid Worksheet figures — Mill Weight (e.g. "485000"), Mill Amount (e.g. "48500"), Warehouse Weight, Warehouse Amount, and a rate field (e.g. "1899.27"), plus an auto-filled reference value — and apply them directly onto the RFQ's estimation data. Saving shows the toast "SEBW values applied to RFQ."

5.3 Bid Performance Tab

Navigation Path: RFQ Master → Bid Performance tab ("/rfq/bid-performance").

A read-only analytical view of win-rate performance. The primary display is a month-by-year pivot table showing, for every month of up to five years (including a selectable year), the bid win ratio both by count of bids (win-ratio-count %) and by dollar value (win-ratio-dollar %), each cell shaded by a three-tier color scale: high (≥ 50%, green), mid (≥ 25%, amber), and low (< 25%, red/neutral). Beneath the table, a year-over-year line/bar chart plots the annual average of both win-ratio metrics so leadership can see whether win rates are trending up or down over time.

5.4 Dollar Dashboard Tab

Navigation Path: RFQ Master → Dollar Dashboard tab ("/rfq/dollar-dashboard").

A financial summary of the entire bid pipeline. At the top, a row of compact stat pills shows TOTAL bids, WON bids, WON in the current year (count and dollar amount), LOST, and PENDING. Below that, a bar chart titled "Bid & Estimation — Total Quoted vs Won / Awarded (by Year)" plots Total Quoted against Won/Awarded dollar value per year — the year for each bid is derived automatically from its Quote Number's two-digit prefix (for example a quote numbered "26-xx-xx" is attributed to 2026). A further monthly chart/table for the current year compares Bid Amount, Awarded Amount, and a per-month Goal figure. The Goal cell for each month is editable inline (an

edit/pencil icon switches the cell into a number input with check/cancel controls) for users with edit rights, and saving shows the toast "Goal updated."

5.5 Job Analytics Tab

Navigation Path: RFQ Master → Job Analytics tab ("/rfq/job-analytics").

A read-only view of historical unit-cost benchmarks: average cost-per-ton and average cost-per-square-foot, pivoted by month and year (covering five years), with computed insight callouts such as the overall average, the lowest cost-per-ton ever recorded, and which job achieved it — useful for sanity-checking a new estimate against historical norms.

5.6 Sales Cycle Tab

Navigation Path: RFQ Master → Sales Cycle tab ("/rfq/sales-cycle").

A "Milestone Day-Counts per Awarded Job" table measuring how many calendar days elapse between key bid milestones (such as quote-to-award) for every awarded job, filterable by year, with summary averages computed across the full data set. The table is paginated (selectable page size) and color-codes long cycle times — values over 365 days are flagged red/danger and values over roughly 180 days amber/warning — to highlight jobs that took unusually long to convert from quote to award.

5.7 Future Capacity Tab

Navigation Path: RFQ Master → Future Capacity tab ("/rfq/capacity").

A rolling 18-month forward projection of fabrication and erection workload, broken out across four trade types — Structural Fabrication, Misc. Fabrication, Structural Erection, and Misc. Erection — each shown with two figures per month: "Potential" hours (the demand from all open bids, excluding any already marked Lost) and "Confirmed" hours (demand from bids that are already Won/awarded). Column totals across all four trades are shown per month. Any month where Confirmed hours exceed a built-in alert threshold of 2,000 hours is visually flagged with a warning icon, giving leadership advance notice of months at risk of being overbooked. This view refreshes itself automatically every five minutes.

5.8 Print Setup Screen

Navigation Path: Reached via "/rfq/print" (a dedicated full-width screen, outside the six-tab layout, generally linked from RFQ Master).

Purpose: Produces a clean, printable "Weekly Meeting View" of upcoming bids for use in physical or PDF hand-outs at internal bid-review meetings.

- **Date range filters:** "Bid Due From" and "Bid Due To" date fields, defaulting to today through 14 days ahead.
- **Results table:** every RFQ whose bid-due date falls in the selected range, with paginated browsing (First/Previous/Next/Last) and a selectable page size.
- **"Print / Save PDF" button:** invokes the browser's native print dialog; the page's print stylesheet automatically hides the header and filter controls so only the clean data table is printed, and the user can choose their browser's "Save as PDF" destination to produce a PDF hand-out.

5.9 Internal Bid Schedule

Navigation Path: Sidebar → BID MANAGEMENT → Internal Bid Schedule ("/bids/schedule").

Purpose: A month-by-month calendar visualization of every RFQ's bid due date, used to plan the team's internal review and preparation workload.

Layout

- **Search box:** placeholder "Search project, quote, customer..."
- **Filters panel:** toggleable filters for Estimator, Decision (to bid), and computed Status.
- **Month navigator:** Previous/Next month icon buttons either side of the displayed month/year heading.
- **Calendar grid:** a full month grid where each day cell lists the bids due that day as small colored chips. Each bid chip is colored by category — "SFE Bid" (decision_to_bid = Yes, blue), "SFE Pre-Bid" (decision_to_bid = TBD, red/pink), or "Other" (amber) — and carries small inline indicators: a status dot (rose = Overdue, orange = Due Today) and a bold "W" or "L" badge if the bid has already been marked Won or Lost.
- **Holiday awareness:** the calendar reads the Holiday Calendar data (Part 5.10) to recognize non-working days when laying out the schedule.
- **"Export" button:** exports the currently filtered set of bids for the visible month to a CSV file named "Bid_Schedule_<Month>_<Year>.csv".

Computed Bid Status

Every bid chip's status badge is computed live from its data: "Won" or "Lost" if already decided; otherwise "No Date" if it has no bid-due date set; "Overdue" if its due date has passed; "Due Today" if the due date is today; or "Upcoming" otherwise.

Action: View Bid Details

- User Action: click any bid chip on the calendar.
- System Result: a details modal opens summarizing that RFQ's key fields without leaving the calendar.

5.10 Holiday Calendar

Navigation Path: Sidebar → BID MANAGEMENT → Holiday Calendar ("/bids/holidays").

Purpose: The central register of non-working days (US Central time zone), consumed by the Internal Bid Schedule and by Capacity Mapping's working-day calculations so that holidays are automatically excluded from available-capacity math.

Who Can Manage: Any signed-in user except the Read-Only role can add, edit, or delete holidays; Read-Only users can view the calendar only.

Layout

- **Year navigator:** Previous/Next year icon buttons either side of the displayed year.
- **Twelve mini month-grids:** one calendar grid per month of the selected year is shown simultaneously. Existing holidays are shown with a solid green day cell; days the user has just clicked to mark (but not yet saved) are shown amber; today's date is highlighted distinctly when otherwise unmarked.
- **Selected-dates editor:** every day the user clicks accumulates in a small pending list, each row letting the user type a Description (placeholder "Holiday description...", required) and choose a Type — Public, Company, or Optional — plus a remove control to drop that date from the pending batch before saving.
- **Existing holidays table:** a list of already-saved holidays for the selected year, each row showing its date, description, and type badge, with inline Edit (pencil) and Delete (trash) icon actions.

Action: Mark and Save New Holidays

- User Action: click one or more day cells across the twelve mini-calendars, fill in a Description (and optionally change the Type) for each in the pending list, then click "Save".

- **Validation:** if any selected date is left without a description, the system blocks the save with the toast "Please add a description for each selected date."
- **System Result:** each valid pending date is created as a new Holiday record; the toast "Holidays saved!" confirms success, the pending list clears, and the new holidays appear as solid green cells and in the table below.
- **Error Scenario:** "Failed to save holidays" (or a server-provided detail message).

Action: Edit an Existing Holiday

- **User Action:** click the pencil icon on a row in the existing-holidays table; the row becomes editable in place (Description text field, Type dropdown) with green checkmark (Save) and grey "X" (Cancel) icon buttons.
- **System Result:** clicking the checkmark saves the change and shows "Holiday updated"; clicking "X" discards the edit.

Action: Delete a Holiday

- **User Action:** click the trash icon on a row, which sets it as the pending delete target and opens a confirmation prompt.
- **System Result:** confirming deletes the record and shows "Holiday deleted"; the row disappears from the table and its calendar cell reverts to unmarked.

5.11 Estimation Summary

Navigation Path: Sidebar → BID MANAGEMENT → Estimation Summary ("/estimation-summary").

Purpose: A searchable, sortable, paginated master list of every project/RFQ that has an estimation record, giving a single place to review past estimates and, where appropriate, kick off a re-bid.

- **Search box:** placeholder "Search projects by name..."
- **Sortable RFQ Date column:** the column header is clickable ("Click to sort by RFQ Date") to toggle ascending/descending order.
- **Pagination controls:** First Page, Previous Page, Next Page, and Last Page icon buttons.

Action: Rebid a Project

- **User Action:** from a project row, the user opens the "Rebid Project" confirmation dialog and confirms.
- **System Result:** the system updates the project/RFQ's status to begin a fresh bidding cycle and shows the toast "Status updated. Loading project for rebid...", then takes the user into the Estimation & Erection module pre-loaded with that project so a revised estimate can be built.
- **Error Scenario:** "Failed to update status for rebid."

PART 6 — ESTIMATION & ERECTION MODULE

This is the heart of the application: the workspace where an estimator turns a won or pending RFQ into a complete, line-item structural-steel cost and weight estimate. It is organized as a project-scoped layout with seven tabs across the top, all operating on the same selected project's estimate. Every numeric field on every tab in this module auto-saves a few moments after the user stops typing (see the Auto-Save Indicator convention in Part 1.6.5), and the entire module becomes permanently read-only for a given project once that project's estimate has been finalized (Part 6.8).

6.1 Module Layout & Project Selection

Navigation Path: Sidebar → BID MANAGEMENT → Estimation ("/estimation-erection"), which defaults to its Estimation Model tab.

At the very top of the layout sits a project selector — a searchable dropdown placeholder "-- Choose Project --" — listing every project/RFQ in the system. Until a project is chosen, every other tab is disabled (each shows the tooltip "Please select a project first" and a centered placeholder message "No Project Selected"). Once a project is chosen, that selection persists as the user moves between the module's seven tabs:

- 10.Contracts — the Contract Information Sheet (Part 6.2).
- 11.Estimation Model — the full numbered bid-cost worksheet (Part 6.3).
- 12.Erection Take-off — crane-pick and field-labor hour estimation (Part 6.4).
- 13.Field Moment Conn. — AISC W-shape moment-connection take-off (Part 6.5).
- 14.Misc Metals — stairs, rails, and other secondary steel items (Part 6.6).
- 15.Breakdown — a cross-module rollup with field-progress annotation (Part 6.7).
- 16.Estimate Data — ERP cost codes, Schedule of Values, and the finalize/lock workflow (Part 6.8).

6.2 Contracts Tab (Contract Information Sheet)

Purpose: Captures the high-level identity, stakeholders, and commercial terms of the project before any cost figures are built.

The tab header shows the project's current status as a colored badge (Planning, In Progress, Yet to Start, or Completed) next to the title "Contract Information Sheet."

Card: Project Profile

- Job Number and Project Name text fields.
- Estimate Date and Project Start Date date fields.
- Sales Representative / PM and Project Manager (PM) — both searchable dropdowns populated from system users.
- Site Location / Address — a multi-line text area.

Card: Stakeholders & Contacts

Captures the Site Foreman (a searchable dropdown of users) and, further down the form, the project's Customer (a searchable dropdown of Customer Master records) along with related contact details inherited from that customer record.

Card: Billing & Commercial Terms

Fields covering payment terms (for example "e.g. Net 45"), and the billing-cycle cut-off days for the month (e.g. "e.g. 20th", "e.g. 25th", or "e.g. End of Month"), plus the state(s) the work will be performed in (e.g. "e.g. PA, NY").

Card: Scope Summary

A free-text area to "Summarize structural scope or key exceptions..." for the project, used as a quick-reference description of what is and is not included in the bid.

6.3 Estimation Model Tab — The Bid Worksheet

Purpose: The primary cost-build worksheet of the entire application: a thirty-step, Excel-aligned calculation that starts from raw material and labor inputs and ends in a single Final Bid Amount.

Header & Controls

- **"Save Draft" button:** saves the estimate with status "in progress." Disabled until a project/RFQ is selected.
- **"Submit" button:** saves the estimate with status "submitted" (the amber/orange primary action).
- **"Clear Form" button:** after the confirmation prompt "Are you sure you want to clear all fields?", resets every input on the worksheet back to blank/zero.
- **Auto-save indicator:** shown whenever a project is selected, reflecting the background auto-save state described in Part 1.6.5.

On a successful Save Draft or Submit, the system shows one of several context-specific success toasts, for example "Draft saved successfully!", "Estimation submitted successfully!", "Project created and estimation submitted successfully!" (when the underlying project record did not exist yet and had to be created from the RFQ), or, if all worksheet totals are exactly zero, "Project reset to Yet to Start!" / "Project created and reset to Yet to Start!". A failed save shows "Failed to save calculations."

KPI Summary Cards

Three live summary tiles sit beneath the header: Total Tonnage, Total Direct Costs, and Final Bid Amount (the last shown with amber emphasis as the headline figure), all recalculating instantly as the worksheet below is edited.

Project Specifications Card

- **PROJECT** — a searchable dropdown of available RFQs/projects ("Select Project"); once a project context is already active (arrived at via Part 5.11's rebid flow or via direct project selection) this field is locked and shows the resolved project code and name, including an "(Unsynced)" suffix for RFQs that have not yet been promoted into Project Master.
- **LOCATION** — free-text site location.
- **MATERIAL DATE** — the date material pricing was last verified, plus further fields capturing the sales representative, project manager, and a project/job number (e.g. "e.g. SFE-2026-904").

The Thirty-Step Bid Calculation Worksheet

The right-hand/lower portion of the Estimation Model tab is a long, numbered worksheet that mirrors a traditional steel-fabrication bid spreadsheet line by line. Every numbered line is a labeled input or a computed roll-up; computed lines are shown read-only with a live-updating value. The thirty steps are organized into five logical phases:

Phase A — Material Direct Costs (Steps 1-9)

- 1. Mill — mill-direct steel weight and dollar amount.

- 2. Warehouse (Whse) — warehouse-stock steel weight and dollar amount.
- 3. Scrap — a scrap percentage applied to material weight/cost.
- 4. Bolts — bolt quantity and unit rate (captured alongside the Scrap section).
- 5. Paint — paint quantity and unit rate.
- 6. Galvanizing — galvanizing weight and unit rate.
- 7. Miscellaneous Items Subtotal — a free-form list of additional named misc material line items, each with its own dollar amount, that roll up into a subtotal.
- 8. Total Material Direct Costs — the computed sum of steps 1–7.
- 9. Material Use Tax — a use-tax percentage applied to the material total.

Phase B — Labor, Shipping & Drafting Direct Costs (Steps 2–13, second numbering pass)

A second labor-and-logistics build-up follows, reusing line numbers 2 through 13 for a separate set of categories:

- 2. Misc Labor — miscellaneous shop labor hours/cost.
- 3. Total Direct Plant Cost — computed shop/plant labor total.
- 4. Freight Out Destination — number of trucks and hours-per-truck for outbound freight.
- 5. Freight Galvanizing — trucks and hours-per-truck for freight to/from the galvanizer.
- 6. Total Shipping Cost — computed roll-up of steps 4–5.
- 7. Sublet Detailing — cost of any detailing work subcontracted out.
- 8. PE Stamp — cost of professional-engineer stamping/certification.
- 9. Total Direct Drafting Cost — computed roll-up of steps 7–8.
- 10. Total Direct Costs — the grand total of the material and labor/shipping/drafting direct costs computed so far.
- 12. Overhead on Direct Costs — an overhead percentage applied to the Total Direct Costs.
- 13. Bid Amount On Direct Costs — Total Direct Costs plus Overhead, i.e. the direct-cost portion of the bid price.

Phase C — Buyout Costs (Steps 21–29)

- 21. Steel Joist and/or Deck — joist/deck buyout cost and tonnage.
- 23. Sublet Erection — cost of any erection work subcontracted out.
- 24. Miscellaneous Metals — buyout cost rolled up from the Misc Metals tab (Part 6.6).
- 25. Other — any other direct buyout cost not covered above.
- 26. Total Direct Buyout Costs — computed sum of the buyout categories.
- 27. Use Tax — use tax applied to the buyout total.
- 29. Total Buyout Costs — Total Direct Buyout Costs plus Use Tax.

Phase D — Buyout Markup (re-using Steps 24–25)

- 24. Buyout Overhead — overhead percentage applied to the buyout total.
- 25. Bid Amount On Buyouts — the buyout portion of the bid price after overhead.

Phase E — Final Roll-Up (Steps 26–30)

- 26. Total Amount Before Profit — Bid Amount on Direct Costs plus Bid Amount on Buyouts.
- 27. Profit — a profit percentage applied to the pre-profit total.

- 28. Total Amount After Profit — pre-profit total plus Profit.
- 29. Misc Charges — any final miscellaneous charges added at the end (for example bonding, permits, or other catch-all fees).
- 30. FINAL BID ESTIMATION AMOUNT — the single headline number representing the complete quoted price for the job, displayed prominently in its own highlighted summary card at the foot of the worksheet, matching the "Final Bid Amount" KPI tile at the top of the page.

Because every rate (overhead %, profit %, scrap %, use-tax %, labor rates, shipping rates, and so on) used throughout this worksheet is configurable per project — driven by values captured on this tab and on the Estimate Data tab — two estimators working on different jobs can apply different commercial assumptions without affecting one another.

6.4 Erection Take-off Tab

Purpose: Estimates the labor hours and equipment cost required to physically hoist and erect the structure on site, based on a piece-count and crane-pick methodology.

Card: Takeoff Configuration Overrides

- Picks Per Day — the assumed number of crane picks the erection crew can complete per working day.
- Crew Size (Onsite) — number of erectors on the crew.
- Foreman Hours Override — manual override for foreman supervision hours, if the computed default needs adjustment.
- Field Studs Quantity — count of shear studs to be field-installed.
- Perimeter Cable (LF) — linear feet of perimeter safety cable required.
- Equipment Rental Cost (\$) — dollar cost of crane/equipment rental.

Grouped Member Tables

Below the overrides, structural members are grouped into category tables (one expandable section per structural category, such as columns, beams, or bracing), each table listing Code, Member Category Description, and Quantity. Within any category, the user can add a custom line item by typing a description and a quantity/unit ("Add custom item description...", a numeric value, and a decimal unit-weight value), and remove any custom item via its trash icon (with the confirmation "Are you sure you want to delete this custom item?"). A separate control lets an estimator define and remove entirely custom units of measure for these line items; removing a custom unit prompts "Are you sure you want to delete the custom unit '<UNIT>'? Any takeoff lines using this unit will be reverted to 'PC'."

Right Sidebar: Erection Summary

A collapsible right-hand panel titled "Erection Summary" totals the take-off, and a further section titled "Erection Self-Perform Labor & Equipment Estimate" presents the resulting computed labor hours and equipment cost for self-performed (non-subcontracted) erection.

Validation: adding a custom item without a description shows "Please enter a description for the custom item." A failed add/delete shows "Failed to add custom item." or "Failed to delete custom item." respectively.

6.5 Field Moment Connections Tab

Purpose: Takes off the bolted/welded field moment-connection scope of the job using standard AISC W-shape steel sections.

Card: Add AISC W-Shape Section

A text field (placeholder "e.g. W14X82, W36...") lets the estimator add a take-off line referencing a standard AISC W-shape designation; the application maintains a built-in reference table of AISC section properties (loaded from the system's AISC section data) to auto-populate the section's engineering properties once selected.

Take-off Lines Table

Each added line can be removed via a trash icon (tooltip "Delete line"). Validation: "Failed to add AISC section line." / "Failed to delete line." on error.

Card: Add Custom AISC Shape

If a required section is not in the built-in AISC reference list, the estimator can define a custom shape (placeholder "e.g. W14X82") with its own properties; the system rejects a duplicate label with "Failed to create custom section. It might already exist."

6.6 Misc Metals Tab

Purpose: Captures secondary/miscellaneous steel scope that falls outside the primary structural take-off: stairs and general accessory items such as handrails, ladders, and gratings.

Section: Stair Flights

- A table of stair-flight line items, each removable via a "Delete flight" trash icon.
- An "Add Stair Flight" builder panel with a Stair Type dropdown ("Select Stair Type") and a Name field (e.g. "e.g. Flight A North"), used to define a new flight's geometry and compute its weight/cost contribution.
- Validation: "Failed to add stair flight." / "Failed to delete stair flight."

Section: Accessories / General Misc Items

- A table of general miscellaneous items (handrails, ladders, gratings, embeds, etc.), each removable via a "Delete item" trash icon.
- An "Add Accessory / Misc Item" builder panel with a Category dropdown ("Select Category") and a description field (e.g. "e.g. Galvanized Platform Grating").
- Validation: "Failed to add miscellaneous item." / "Failed to delete miscellaneous item."

A right-hand "Misc Metals Summary" panel totals the tonnage and cost of everything entered on this tab, which then feeds into Step 24 (Miscellaneous Metals) of the Estimation Model worksheet (Part 6.3).

6.7 Breakdown Tab

Purpose: A cross-module rollup view of the estimate broken into its component line items, doubling as a lightweight field-progress tracker once fabrication begins.

Breakdown Matrix Table

Lists every component of the estimate with its Component Description, Weight (Tons), Shop Hrs, Field Hrs, Valuation (\$), and a Status badge — ESTIMATED (default, grey), PROCURED (blue, once material has been ordered), or ERECTED (green, once the component has been physically erected). A "Galv" tag appears next to galvanized components. Clicking any row selects it (highlighted amber) and opens the Tracking Control Drawer on the right.

Tracking Control Drawer

For the selected row, this drawer lets the user record a Detailer / Engineer Assigned (free text, e.g. "e.g. John Doe detailing"), Notes / Revisions (a multi-line field for field revision notes), and two progress checkboxes — Material Ordered / In Shop and Erected — that directly drive the row's Status badge in the matrix table above. Every change in this drawer auto-saves.

6.8 Estimate Data Tab — Cost Codes, SOV & Finalization

Purpose: Surfaces the estimate's data in standard ERP cost-code and Schedule-of-Values formats, supports exporting the estimate, and houses the formal finalize/lock (sign-off) workflow.

Card: 14 ERP Cost Codes

A read-only table mapping the estimate's totals onto the company's fourteen standard ERP cost codes, with a computed grand total shown at the foot of the table.

Card: Schedule of Values (SOV)

A table of the seven standard SOV line items the system seeds automatically for every project — Bond (code 50), Drawings (100), Steel (200), Joist (300), Deck (400), Miscellaneous Metals (500), and Erection (600) — each with an editable Amount field that auto-saves on change, plus an SOV grand total. Additional custom SOV lines can be present alongside the seeded seven.

Card: Estimate Approval

- Prevailing Wage Job and Out-of-Town Job — two checkboxes capturing commercial flags relevant to the bid.
- Finalize control: while the estimate is not yet finalized, a "Finalize Estimate" button (lock icon) is shown; once finalized, this is replaced with a disabled, greyed-out "Estimate Locked" indicator.

Action: Finalize (Lock) the Estimate

- User Action: click "Finalize Estimate".
- System Result: a confirmation modal titled "Finalize Estimate Submission" appears, warning: "CONFIRMATION REQUIRED: Finalizing this estimate will generate an immutable snapshot (v<next version number>) and freeze all inputs in this workspace. Corrections will require cloning to a new job revision." The user must type their full name into a "Preparer Name Signature" field (acting as a digital signature) before the form can be submitted.
- On confirming ("Lock Estimate"), the system records the preparer name and the Prevailing Wage / Out-of-Town flags, creates a new immutable revision snapshot, and marks the project's estimate as finalized — at which point every tab across the entire Estimation & Erection module for that project becomes read-only.
- Clicking "Abort" closes the modal without finalizing.
- Error Scenario: "Failed to finalize estimate: <server detail, e.g. Conflict>."

Card: Revision History

Lists every prior finalized snapshot ("Revision v<n>") for the project, each showing its finalized date and the preparer's name, plus per-snapshot "PDF" and "XLSX" download links that export that point-in-time estimate. If no snapshots exist yet, the panel reads "No snapshots registered for this job."

PART 7 — BUDGET ESTIMATOR MODULE

The Budget Estimator is a lightweight, fast "ballpark" weight-and-cost calculator, separate from the full line-item build-up performed in the Estimation & Erection module (Part 6). It is meant for quick early-stage budget numbers — for example to give a client a rough \$/ton figure before a detailed estimate is built — based on building square footage and a handful of weight-per-square-foot assumptions. It consists of two sidebar entries, Design Inputs and Estimation Result, which work together against the same project record (its results are stored on the project's "steel_budget_worksheet" field).

7.1 Design Inputs Screen

Navigation Path: Sidebar → BUDGET ESTIMATOR → Design Inputs ("/steel-budget/input").

Purpose: Lists every project that already has quick-budget design inputs saved, and is the entry point for creating or editing those inputs.

- **"Add Inputs" button:** opens the Design Inputs Wizard modal (described below) to start a new quick-budget worksheet for a project.
- **Project list:** shows only projects that already have a non-empty steel_budget_worksheet saved, each with View, Edit (pencil), and Delete (trash) row actions.

Action: View

Clicking the eye/View icon stores that project as the "active budget project" and navigates straight to the Estimation Result screen (Part 7.2) to show its computed output.

Action: Edit

Clicking the pencil/Edit icon re-opens the Design Inputs Wizard pre-filled with that project's existing worksheet values.

Action: Delete

- **User Action:** click the trash icon; the system prompts "Are you sure you want to delete the design inputs for project <code>? This will also remove the estimation results."
- **System Result:** on confirm, the project's steel_budget_worksheet is cleared and its total_ton value reset to zero; the row disappears from this list.
- **Error Scenario:** "Failed to delete design inputs. Please try again."

7.1.1 The Design Inputs Wizard

A three-step modal wizard (with a progress bar showing "Step <n> of 3") used to both create and edit quick-budget worksheets.

Step 1 — Project Areas

- **Select Project** — a dropdown choosing which project this worksheet belongs to (locked/disabled once editing an existing entry).
- **An area table** with one row per building level — Level 1, Level 2, Level 3, Level 4, and Roof — each row capturing an Area (SF) value, an "Include in Steel?" checkbox (excluding a level from the steel-weight math if unchecked), and a free-text Notes field.
- **Total Included Floor SF** — a computed, read-only sum of every level marked "Include in Steel."
- **Roof SF** — a separately editable roof-area figure.

Step 2 — Steel Weight Assumptions

Eight weight/percentage assumption fields used to convert square footage into an estimated steel tonnage:

- Typical Floor Framing (lbs/SF) — e.g. "15.0".
- Roof Framing (lbs/SF) — e.g. "12.0".
- Misc/Secondary Steel (lbs/SF) — e.g. "2.5".
- Moment Frame Adder (%) — e.g. "1.0".
- Façade Support Premium (%) — e.g. "3.0".
- Rooftop Allowance (lump sum lb) — e.g. "15000".
- Canopy Allowance (lump sum lb) — e.g. "25000".
- Contingency (%) — e.g. "5.0".

Step 3 — Costing

- Budget \$/Ton (enter 0 to ignore) — a dollar-per-ton rate, e.g. "5500", used to convert the computed tonnage into a dollar budget.
- A live "Estimated Output Preview" card shows the resulting Est. Steel Tons and Total Budget figures, recalculated instantly as inputs change, before the user even saves.

Wizard Navigation

- "Back" — returns to the previous step (disabled on Step 1).
- "Next Step" — advances to the next step (shown on Steps 1 and 2).
- "Add Save Inputs" — shown only on Step 3; saves the worksheet to the project. Shows a spinner and "Saving..." while in flight.

Validation: if no project has been selected, attempting to save shows "Please select a project first." A failed save shows "Failed to save inputs. Please try again."

7.2 Estimation Result Screen

Navigation Path: Sidebar → BUDGET ESTIMATOR → Estimation Result ("/steel-budget/result"); also reached directly via the View action on the Design Inputs screen.

Purpose: Displays — and allows manual override of — the computed weight summary, cost summary, and contingency premiums for a project's quick-budget worksheet.

- **Project dropdown selector:** lets the user switch which project's results are shown; the last-viewed project persists across navigation via the browser session.
- **Result panel:** renders the same worksheet fields as the wizard but laid out for review, including the recalculated total tonnage and total dollar budget; values here can be directly edited as manual overrides on top of the original wizard inputs.
- **"Save Results" button:** persists any manual overrides made on this screen back to the project record. Shows a spinner and "Saving..." while in flight; on success shows the alert "Steel budget estimation results saved successfully!"; on failure, "Failed to save estimation results. Please try again."

If no project is selected, the screen shows a placeholder state: "No Project Selected — Please choose a project from the dropdown in the header to view or override Estimation Results."

PART 8 — MASTER DATA MANAGEMENT

This Part covers the three "Master" directory screens, grouped under the Sidebar's MANAGEMENT section, that hold the reference data referenced throughout the rest of the system: Employee Master, Customer Master, and Detailer Master. All three follow the same overall pattern — a filterable table with Add/View/Edit/Delete actions opening a modal form — described once here with each screen's specific fields noted.

8.1 Employee Master

Navigation Path: Sidebar → MANAGEMENT → Employee Master ("/employees").

Purpose: The directory of all company personnel — shop staff, estimators, project managers, and administrators — referenced by searchable dropdowns throughout the system (for example as Project Manager or Site Foreman choices).

Toolbar & Filters

- "Add Employee" button — opens the registration modal in create mode.
- Three searchable-dropdown filters: Employee name ("All Employees"), Designation ("All Designations"), and Department ("All Departments" — Fabrication, Design, Quality, Admin, or Operations).

Table Row Actions

- View (eye icon) — opens the same modal in read-only mode.
- Edit (pencil icon) — opens the modal pre-filled and editable.
- Delete (trash icon) — prompts "Are you sure you want to delete this employee?"; on confirm, deletes the record. Error: "Failed to delete employee."

The Employee Form — A Two-Step Wizard

The Add/Edit/View modal is a two-tab wizard, "Personal" then "Employment," navigated with Next Step / Previous controls.

- **Step 1 — Personal:** Full Name, Email Address, Phone Number (must be exactly 10 digits — the field strips any non-digit characters as the user types and the browser enforces a 10-character pattern), Location, Date of Birth, and Gender (Male / Female / Other).
- **Step 2 — Employment:** Employee ID (e.g. "SFE-000"), Designation (e.g. "Project Manager"), Department (Fabrication / Design / Quality / Admin / Operations), and Current Status (Active / On Leave / Inactive).

On the Personal step, clicking "Next Step" simply advances to Employment without saving. The final submit button on the Employment step reads "Create Identity" when adding a new employee, or "Update Profile" when editing an existing one, and is hidden entirely in View mode (replaced by a single "Close" button). Validation: a failed save surfaces every field-level error message returned by the server, concatenated into one alert.

8.2 Customer Master

Navigation Path: Sidebar → MANAGEMENT → Customer Master ("/customers").

Purpose: The directory of client companies that issue RFQs; referenced by the Customer dropdown on RFQ and Project records.

Toolbar & Filters

- "Add Customer" button.

- Three searchable-dropdown filters: Customer name ("All Customers"), Code ("All Codes"), and Country ("All Countries").

The Customer Form

- Customer Name, Customer Code, Country (a dropdown — India, USA, UK, UAE, or Other), Street, State, Address, and Designation.
- **Multiple Contacts:** a repeatable contact block (Contact Person Name, Email Address, Phone) that can be added or removed per customer; at least one contact row is always present by default. Each contact's phone field auto-prefixes the dialing code matching the selected Country (for example "+91 " for India, "+1 " for USA, "+44 " for UK, "+971 " for UAE); changing the Country re-prefixes every existing contact's phone number accordingly. A "Delete Contact" icon removes an individual contact row.

Row Actions

- View, Edit, and Delete icons behave exactly as in Employee Master.
- Delete confirmation: "Are you sure you want to delete this customer?"; error: "Failed to delete customer."
- Save error: "Failed to save customer."

8.3 Detailer Master

Navigation Path: Sidebar → MANAGEMENT → Detailer Master ("/detailers").

Purpose: The directory of external structural-detailing partners/vendors who translate the awarded design into shop and erection drawings; referenced by the Detailer dropdown on Project Master.

Toolbar & Filters

- "Add Detailer" button.
- Two searchable-dropdown filters: Detailer name ("All Detailers") and Code ("All Codes").

The Detailer Form

Detailer Name and Detailer Code, plus the same repeatable Contacts block used on Customer Master (Contact Person, Email, Phone with country-aware dialing-code prefix and a "Delete Contact" control).

Row Actions

View, Edit, and Delete icons behave the same way as the other two masters. Delete confirmation: "Are you sure you want to delete this detailer?"; errors: "Failed to delete detailer." / "Failed to save detailer."

PART 9 — PROJECT MANAGEMENT MODULE — PROJECT MASTER

Navigation Path: Sidebar → PROJECT MANAGEMENT → Project Master ("/projects").

Purpose: The master system-of-record for every active or completed structural-steel project. Beyond basic project identity, every project also owns a detailed Structural Schedule — a sequence-by-sequence breakdown of the job's tonnage and key dates — captured and edited from a second tab inside the same Add/Edit modal.

9.1 Project List Screen

Control Bar

- Search box — placeholder "Search projects by name, code or customer..."
- Status filter dropdown — All / In Progress / Yet to Start / Completed.
- "Export CSV" button — exports the filtered project list.
- "Add Project" button — opens the project modal in create mode, defaulting straight to its Basic Details tab.

Table Columns

Project Name · Code · Customer Name · Detailer Name · Manager · Erection (date) · Priority (color badge: red=High, amber=Medium, green=Low) · Ton · MH's (total man-hours) · MH's/Ton (computed) · Actions.

Row Actions

- **Structural Plan (document icon):** generates and downloads a formatted PDF of the project's structural schedule/Gantt plan.
- **View (eye icon):** opens the project modal in read-only mode.
- **Edit (pencil icon):** opens the project modal in edit mode.
- **Delete (trash icon):** prompts "Are you sure you want to delete this project?"; on confirm, removes the project (and, on the backend, cascades to delete its associated structural-schedule items and any related production-schedule items). Error: "Failed to delete project."

The table is paginated (10 rows per page) and shows a loading spinner while data is being fetched, or "No projects found matching your search." when the filtered list is empty.

9.2 The Project Modal — Basic Details Tab

When creating a new project, an optional "Pre-fill Project from Won RFQs" panel offers a searchable dropdown of every RFQ marked Won; selecting one auto-populates Project Name, Project Code (preferring the RFQ's assigned SFE Job # if one exists, otherwise its Quote No), Customer Name, Total Ton, and Total Manhours from that RFQ's data.

Fields

- Project Name and Project Code (e.g. "PRJ-001") — free text.
- Customer Name — read-only, shown as "Synced from RFQ" since it is only ever set via the RFQ pre-fill above.
- Detailer Name — a searchable dropdown of Detailer Master records.
- Project Manager — a searchable dropdown limited to Employee Master records whose Designation is "Project Manager."

- Erection Date — once the project is being edited (not on initial creation), this becomes a read-only display rather than an editable field.
- Project Priority — Low / Medium / High.
- Project Status — In Progress / Yet to Start / Completed.
- A highlighted "Total Ton / Total Manhours / Manhour per Ton" block — Total Ton and Total Manhours are editable; Manhour per Ton is auto-computed ($\text{Total Manhours} \div \text{Total Ton}$) and updates live, which in turn also recalculates every Structural Schedule row's Budget Plant Hours figure ($\text{tons} \times \text{the new manhour/ton rate}$) automatically.
- A second highlighted block for Plant Name (plant1 / plant2 / plant3) and "Is Scheduled Field Measure Date Required?" (Yes/No) — when set to "No," the Scheduled Field Measure Date column is hidden entirely from the Structural Schedule grid on the second tab for this project.

Validation: Project Name and Project Code are both required; attempting to save without them shows "Project Name and Code are required." A failed save shows the server-reported error detail.

9.3 The Project Modal — Structural Schedule Tab

The second tab, "Structural Schedule," switches the modal to a wide, spreadsheet-style grid (the modal itself widens to nearly full-screen for this tab) listing every sequence of the project. A "Download PDF" button appears in the modal header only while this tab is active.

" + Add Sequence" Button

Appends a new, blank sequence row to the grid (not yet saved to the server until the modal is saved), pre-numbered with the next sequence number.

Grid Columns

Every row represents one structural sequence and is directly editable cell-by-cell:

Column	Description
SEQ #	The sequence identifier within the project (e.g. "1", "2A").
Tons	The tonnage of steel in this sequence.
Item Description	Free text describing the sequence, plus two small buttons: "Add Item" (enabled only for newly added, unsaved rows) and "View" (shows a count of fabrication detail line items if any exist) — both open the Fabrication Details sub-modal (Part 9.4).
Scheduled OFA Date	The planned date Originals-For-Approval drawings are issued.
Scheduled BFA Date	The planned date Back-For-Approval (approved/returned) drawings are received.
Scheduled Field Measure Date	The planned date for field measurement (hidden entirely if the project's "Schedule Field Measure Required" flag is set to No).
RTS Date	The planned Ready-To-Ship date — the single most important scheduling date, used throughout Dashboard, Plan Tracking, and Production for status/health calculations.
Plant Lead Time in Weeks	How many weeks of shop lead time this sequence requires.
Scheduled Start of Erection	The planned date field erection of this sequence begins.
Budget Plant Hours	The budgeted shop man-hours for this sequence (auto-calculated from the project-level manhour/ton rate, but manually overridable).
Budget Field Hours	The budgeted field-erection man-hours for this sequence.

Column	Description
Plant Hours Actual	Actual shop hours logged so far.
Field Hours Actual	Actual field hours logged so far.
Detailer / Vendor	Which detailing partner is responsible for this sequence.
Dwg Status	Free-text drawing status note.
Notes	Free-text notes.
Action	A trash icon to remove the row from the grid (removed rows are deleted from the server when the modal is saved).

A header strip above the grid restates the project's Customer, Job #, PM, Detailer, Erection Date, Tons, MH's, and MH/Ton for quick reference while editing.

9.4 Fabrication Detail Sub-Modal

Purpose: Captures the granular bill-of-material detail behind a single structural-schedule sequence — useful for shop planning once a sequence reaches detailed fabrication.

Opened via the "Add Item" or "View" buttons on any Structural Schedule row. Presents a table with columns Pieces (quantity), Material Size (e.g. "e.g. W14x22"), Dimension (e.g. "e.g. 20'-0\""), and Machine (the shop machine assigned to cut/process this material). An "Add More Items" button appends additional blank rows, and each row has its own delete (trash) icon. "Save Details" commits the (non-empty) rows back onto the parent sequence's fabrication_details list; "Cancel" discards changes.

PART 10 — STRUCTURAL SCHEDULE MODULE

The Structural Schedule module gives a sequencing-focused view of the exact same project and structural-sequence data captured in Project Master (Part 9), through two dedicated screens: Plan Creation (for setting up and visualizing the sequence plan) and Plan Tracking (for recording actual progress against that plan).

10.1 Plan Creation

Navigation Path: Sidebar → STRUCTURAL SCHEDULE → Plan Creation ("/structural/plan-creation").

Purpose: A project list, scoped specifically to defining and visualizing each project's structural sequencing plan, with an inline Gantt-chart preview per project.

Toolbar

- Search box — placeholder "Search projects..."
- "Create New Plan" button — opens the same two-tab project modal described in Part 9.2–9.3 (Basic Details + Structural Schedule), used here to define a brand-new project's sequencing plan from scratch.

Table Columns

Project Name · Code · Customer · Erection Date · Actions.

Row Actions

- **Structural Plan (document icon):** downloads a formatted PDF of the project's sequence plan, named "StructuralPlan_<project code>.pdf".
- **Toggle Gantt (eye icon):** expands an inline Gantt-chart row directly beneath that project, plotting every one of its structural sequences on a timeline (with field-measure milestones shown only if the project requires them). Clicking the icon again collapses it.
- **Edit (pencil icon):** opens the project modal pre-loaded with that project's Basic Details and full Structural Schedule grid for editing.
- **Delete (trash icon):** prompts "Are you sure you want to delete this project?"; on confirm, deletes the project. Error: "Failed to delete project."

Validation when saving a plan: Project Name and Project Code are required ("Project Name and Code are required"); a failed save shows the detailed server error.

10.2 Plan Tracking

Navigation Path: Sidebar → STRUCTURAL SCHEDULE → Plan Tracking ("/structural/plan-tracking").

Purpose: The day-to-day working screen for recording actual progress — actual approval, ready-to-ship, and erection dates — against every project's planned structural sequence, alongside a live Gantt visualization of both planned and actual timelines.

Search

A single search box (placeholder "Search projects or tracking items...") filters by project name, code, customer, or, once a specific project is opened, by its sequences' description/category/notes.

Project Selection & Sequence Table

Selecting a project from the filtered list reveals all of its structural sequences in a tracking table. Each sequence row shows its planned dates alongside editable Actual Date fields; entering an actual date for a milestone (for example Actual RTS Date) immediately saves it to the server.

Validation: actual dates cannot be set in the future — attempting to enter a future date shows "Actual dates cannot be set in the future." and the edit is rejected. A failed save shows "Failed to update <field name>."

Outlook Completion Date

For each sequence, the table also calculates and displays an "Outlook Completion Date" — the planned RTS date plus that sequence's Plant Lead Time in Weeks — giving a forward projection of when shop work on that sequence should wrap up, independent of whether an actual date has been recorded yet.

Gantt Visualization

A Gantt-chart component renders every visible sequence as a horizontal timeline bar, letting a user see at a glance which sequences are on schedule, completed, or behind, across all open projects or within a single selected project.

PART 11 — PRODUCTION MANAGEMENT MODULE

Production Management is where structural-schedule sequences are released to the shop floor and routed through the fabrication process. It consists of "Production Schedule" — itself a two-tab screen holding the Production Priority Schedule and Process Master — plus a separate Process Master Settings screen used to configure per-routing-station rates.

11.1 Production Schedule (Two-Tab Layout)

Navigation Path: Sidebar → PRODUCTION MANAGEMENT → Production Schedule ("/production"), which defaults to the Priority Schedule tab.

A thin tab bar lets the user switch between "Production Schedule" (the Priority Schedule screen, Part 11.2) and "Process Master" (Part 11.3); only the tabs the user's role/permissions allow are shown.

11.2 Production Priority Schedule

Purpose: Groups structural-schedule sequence work into named "production schedules," each spanning a start and end date, and prioritizes/sequences the underlying job items for shop release.

Toolbar

- Search box — placeholder "Search by Schedule #..."
- "Filters" button.
- "Export CSV" button.
- "New Schedule" button — opens the schedule modal.

List

Each row is one production schedule: Schedule Number, Production Start Date, Production End Date, and Actions. If no schedules exist yet, the screen shows the empty-state message "No production schedules yet — Click \"New Schedule\" to start defining your production priorities."

Row Actions

- (Row click) — expands an inline table beneath the row listing every production item assigned to that schedule.
- **View Gantt Chart (eye icon):** switches the expanded inline view to a Gantt-chart visualization of the schedule's items instead of the table, with a timeline-range filter (All / 1 Month / 3 Month / 6 Month).
- Edit (pencil icon) — opens the schedule modal pre-filled for editing.
- Delete (trash icon) — prompts "Are you sure you want to delete this schedule and all its items?"; error on failure: "Failed to delete schedule."

The Schedule Modal

- Schedule Number, Start Date, and End Date — all required.
- A search box (placeholder "Search by code or name...") to find and add specific structural-schedule items/sequences into this production schedule.
- A line-item table where each added item shows its Job # (read-only, placeholder "Job #"), computed weight figures, an editable Notes field (placeholder "Add notes..."), and numeric fields for quantities/values.

Validation on save: "Save Failed: Schedule Number is required.", "Save Failed: Start Date is required.", "Save Failed: End Date is required.", or "Save Failed: Please add at least one production item with a Job #" if the item table is empty.

Gantt Bar Color Coding (within the expanded Gantt view)

Each item's bar is colored from its computed timeline status: green ("Completed," once a ship date is recorded), blue ("In Progress," its planned RTS date is today), yellow ("Likely Delay," 1–7 days past the planned RTS date with no actual date yet), red ("Delayed," more than 7 days past), or slate ("Yet to Start" / "TBD" when no RTS date is set at all) — the same adherence logic used elsewhere in the system (Part 1.6.6, Part 4.2.2).

11.3 Process Master

Purpose: Records, per production schedule, exactly which routing/process stations each job's items have been assigned to, across three priority categories — Plate, Angle, and Structural.

Toolbar

- Search box — placeholder "Search projects by name, code or schedule..."
- "Add Process Station" / new-record entry button, opening the Process Master form.

Row Actions

Each row (grouped by production schedule) offers View, Edit, and Delete icon actions; Delete prompts "Delete all process records for schedule \"<schedule number>\"?" and on failure shows "Failed to delete one or more records. Please try again."

The Process Master Form — A Guided Wizard

A multi-step wizard ("BASIC INFO" first, then one step per priority category) lets the user assign items to routing stations:

- **Basic Info step:** select the production schedule this entry belongs to.
- **Plate Priority step:** assigns items to the Plate routing stations — Plasma, Plate Over -1, and Bent Plate.
- **Angle Priority step:** assigns items to the Angle routing stations — Angle Master, Ironworker, and Peddinghaus (Large).
- **Structural Priority step:** assigns items to the Structural routing stations — Peddinghaus Drill Line, Ficep, and Punch.

Within any step, each line item carries a fixed set of columns: Job #, Seq #, Weight (in Tons), Weight (in LBS, read-only computed), RTS Date (read-only), Actual RTS Date (editable), Run Days (read-only computed), Start Run Date (read-only computed), Complete Run Date (read-only computed), and Notes. Validation: "Please select a schedule" if no schedule is chosen before saving; "Failed to save data." on a server error.

11.4 Process Master Settings

Navigation Path: Sidebar → PRODUCTION MANAGEMENT → Process Master Settings ("/production/master-settings").

Purpose: Configures the per-process-station billing/production rate used when items are routed through Process Master, for a selected production schedule.

If no production schedule exists yet anywhere in the system, the screen automatically creates a default one named "SCH-01" so rate configuration can begin immediately. A schedule selector lets the user switch which schedule's rates are being edited.

The Nine Configurable Process Stations

Grouped into the same three priority categories used in Process Master, each station has an editable Rate field (defaulting to the values shown):

Priority Category	Stations & Default Rates
Plate Priority	Plasma (4500) · Plate Over -1 (2000) · Bent Plate (3000)
Angle Priority	Angle Master (0) · Ironworker (0) · Peddinghaus (Large) (0)
Structural Priority	Peddinghaus Drill Line (0) · Ficep (0) · Punch (0)

A "Save" action persists the edited rates for the selected schedule. On success: "Process master settings saved successfully!" On failure: "Failed to save process master settings." If no schedule is available to save against: "No schedule available to save settings."

PART 12 — CAPACITY MAPPING MODULE

Navigation Path: Sidebar → CAPACITY MAPPING → Capacity Configuration / Machine Master / Workforce Master (all share the base route `/production/capacity-mapping/<tab>`).

Purpose: Cross-checks the shop's actual machine and labor capacity against the production demand generated by every Won bid, so leadership can see, by shop/location and by month, whether the company has enough capacity to deliver its current pipeline.

All three sidebar entries land on one shared screen with a persistent "Performance Overview" summary at the top and three switchable tabs (Capacity Configuration, Machine Master, Workforce Master) below it, selected via the URL's tab segment.

12.1 Performance Overview (Shared Header)

Filters

- **Filter by Month:** a checkbox dropdown ("All Months" plus each individual month) allowing a multi-month selection; if nothing is explicitly selected, the view defaults to the current month.
- **Filter by Shop:** a dropdown of every distinct shop/location value found in the Capacity records, plus "All Shops".

The Six KPI Cards

Card	What It Shows
Total Capacity	Total tonnes-per-month capacity across all Capacity Configuration records (filtered by the selected shop).
Manhours Available	Total monthly man-hours sourced from the Workforce Master records active in the selected month(s).
Manhours Required	Computed demand: the sum of structural/misc fabrication and erection hours from every Won RFQ bid that is scheduled to be active during the selected month(s), based on each bid's start month and duration fields.
Surplus/Shortage Manhours	Manhours Available minus Manhours Required. Shown in green as "Surplus Manhours" when positive ("We have more manhours than required"), or in rose/red as "Shortage of Manhours" when negative ("We need more manhours").
Total Machinery	Count of machines registered in Machine Master.
Total Manpower	Count of skilled-manpower records registered in Workforce Master.

12.2 Capacity Configuration Tab

Purpose: Defines the company's raw production capacity, per shop/location and per process, in both machine-driven and manual terms.

"New Capacity Entry" Form

- Shop and Location — free text (e.g. "USA, India").
- Category — Machine or Manual.
- Process — required free text (e.g. "Drilling").
- An optional link to a specific Machine record (when Category = Machine).

- Capacity figures — per-day, with per-month and per-year values either entered directly or automatically derived from the per-day rate multiplied by the number of working days in the selected month(s)/year, taking the Holiday Calendar into account so holidays and weekends are excluded from the working-day count.
- Rate Per Day and a Months selector — which months of the year this capacity entry applies to.

Validation/Errors: "Failed to save capacity." on save failure. Delete prompts "Delete this capacity record?"; failure shows "Failed to delete."

12.3 Machine Master Tab

Purpose: The equipment register for every production machine in the shop, also used to populate the Machine dropdown when linking a Capacity Configuration entry to specific equipment.

"Register Machine" Form

- Machine Name (required, e.g. "e.g. CNC Plasma Cutter") and Machine ID (e.g. "MC-001").
- Make (required, e.g. "e.g. Voortman"), Model No (e.g. "V630"), and Serial No (e.g. "SN-12345").
- Shop (e.g. "e.g. Shop 1") and a process/department label (e.g. "e.g. Fabrication").
- Validity Year (e.g. "12/26") and Capacity Per Day.
- A free-form set of additional custom Label/Value attribute pairs that can be added for machine-specific specifications not covered by the fixed fields above.

Required-field validation highlights the Name and Make fields in red if left blank. Save failure: "Failed to save machine." Delete prompts "Delete machine?" directly from the table row.

12.4 Workforce Master Tab

Purpose: The labor register used to compute monthly Manhours Available in the Performance Overview, broken out by skill level and process.

Fields (driven by the Manpower data model)

- Employee Name and Process (the task/skill this entry covers).
- Skill Level — High, Medium, or Low.
- Productivity Rate Per Day and Rate Per Day.
- Manhours — defaults to 8.00 per day — and Overtime hours.
- Month — which single month this manpower allocation applies to (January through December).

Save failure: "Failed to save manpower."

PART 13 — SYSTEM MODULE

The SYSTEM section of the Sidebar groups three screens that are not tied to a specific project or bid: Settings (personal account preferences), Announcements (company-wide notices), and User Access (administrator-only account and permission management).

13.1 Settings

Navigation Path: Sidebar → SYSTEM → Settings ("/settings").

Purpose: Lets the signed-in user manage their own profile, password, and display preferences.

The screen is organized into three tabs: My Profile, Security, and Appearance.

My Profile Tab

- First Name, Last Name, Email Address — editable text fields.
- Job Role — a dropdown (Administrator / Project Manager / Employee) shown for display purposes.
- Department — shown but disabled (read-only).
- "Save Changes" persists the profile; on success the cached session user record is updated immediately so the Sidebar and Header reflect the new name/email without a page reload.

Validation: a failed save surfaces the first field-level error returned by the server, or the fallback "Failed to update profile."

Security Tab

- Current Password, New Password, and Confirm New Password — all required, masked fields.
- "Update Password" submits the change.
- Validation: "New passwords do not match." if the two new-password fields differ.
- Error: the server's specific old-password error, or "Failed to update password."
- **Danger Zone:** a "Delete Account" button which, after the confirmation prompt "Are you absolutely sure you want to delete your account? This action cannot be undone.", permanently deletes the signed-in user's own account, clears the session, and returns the browser to the Login screen. Failure shows "Failed to delete account. Please contact support."

Appearance Tab

- Theme Preferences — Light Mode or Dark Mode selectable as two large preview cards; the choice is saved to the browser and applied immediately.
- System Language — a dropdown (English (US) or Hindi).
- Timezone — a dropdown (Asia/Kolkata or UTC).

13.2 Announcements

Navigation Path: Sidebar → SYSTEM → Announcement ("/announcements").

Purpose: Lets Admins/Managers publish company-wide notices that appear in the Dashboard's "Internal Notices" panel (Part 4.2.7) and, while active, on the Login screen for all users.

Create / Edit Form

- Title and Message (multi-line) — both required.

- From Date and To Date — both required; an announcement is only considered "Active" within this date window.
- Active (visible on login page) checkbox — defaults checked.
- "Publish Announcement" (create) / "Update Announcement" (edit) submit button, with a "Cancel" button shown only while editing.

Validation: "Please fill in all required fields." if Title, Message, From Date, or To Date is missing. Success: "Announcement published successfully." or "Announcement updated successfully." Failure: the server-reported detail, or "Failed to save announcement. Please check your data."

Existing Announcements List

Every announcement is shown as a card with its Title, a status badge (Active / Inactive / Expired — computed by comparing today's date against the To Date and the Active flag), creation date, message body, its active date range, and the name of the user who created it.

Row Actions

- **Activate / Deactivate:** toggles the Active flag; shows "Announcement activated successfully." or "Announcement deactivated successfully."
- **Edit:** loads the announcement back into the form above for editing (the page scrolls to the top).
- **Delete:** prompts "Are you sure you want to delete this announcement?"; on confirm shows "Announcement deleted successfully." Failure: "Failed to delete announcement."

13.3 User Access (Administrator Only)

Navigation Path: Sidebar → SYSTEM → User Access ("/user-access") — visible only to users with the Admin role; any other role attempting to reach this URL directly is redirected away.

Purpose: The single screen for creating new user accounts, assigning roles, and controlling exactly which of the twenty application modules each individual user is permitted to see (the `allowed_modules` list referenced throughout Part 1.5.2).

Toolbar

- Search box — placeholder "Search by name, email, role, initials..."
- A role filter.
- A CSV export of the visible user list.
- "Register New Account" button — opens the registration modal.

User Table & Inline Module-Access Grid

Each user row shows their identity (avatar/initials, name, email), role badge, and a grid of per-module toggle checkboxes spanning all nine Sidebar categories (Overview, Bid Management, Management, Budget Estimator, Project Management, Structural Schedule, Production Management, Capacity Mapping, System). Clicking any module checkbox for a user immediately calls the server to add or remove that module key from the user's `allowed_modules` list — there is no separate "Save" step for this grid.

- Success: "Access updated for <user name>."
- Failure: "Failed to update module access permissions."

Row Actions

- View Details (eye icon) — opens a read-only "User Account Details" modal summarizing the account.

- Edit User (pencil icon) — opens the "Edit User Profile" modal.
- Delete User (trash icon) — opens a "Confirm Delete User" modal.

Action: Register a New Account

The "Register New Account" modal collects: First Name (e.g. "e.g. John"), Last Name (e.g. "e.g. Doe"), Email (e.g. "e.g. john.doe@steelfab.com"), Initials (e.g. "e.g. JD"), Role (Admin / Manager / Estimator / Employee / Detailing-JF), Department (e.g. "e.g. Estimation"), Phone (e.g. "e.g. 555-0199"), Password (placeholder "Min. 6 chars"), and Confirm Password (placeholder "Repeat password"). An "Employee" picker can also be used to auto-fill the Name/Email/Initials/Department/Phone fields directly from an existing Employee Master record, to keep the two directories consistent.

- Validation: a valid email is required; First Name and Last Name are required; Initials must be 2–5 characters; Password is required and must be at least 6 characters; Confirm Password must match.
- Success: "User registered successfully."
- Failure: field-specific errors returned by the server are shown next to each field, or the fallback toast "Failed to register user."

Action: Edit a User

The same field set as registration, minus the password requirement (a password is only validated/changed if the user chooses to enter a new one in the New Password / Confirm New Password fields). Success: "User updated successfully." Failure: field-specific errors, or "Failed to update user details."

Action: Delete a User

The "Confirm Delete User" modal requires an explicit confirmation click before the account is permanently removed. Success: "User deleted successfully." Failure: the server's detail message, or "Failed to delete user."

Action: Export Users to CSV

Exports the currently visible user list to CSV; if there is no data to export, shows "No user data to export," otherwise "CSV exported successfully."

PART 14 — AI CHATBOT ASSISTANT — "STEEL FAB ASSIST"

This Part describes the very assistant that this document is written to feed. Understanding how it works helps explain both what it can answer and the boundaries of what it can do.

14.1 How the Assistant Is Grounded (RAG Architecture)

The assistant does not rely purely on a language model's general knowledge. Instead, it uses Retrieval-Augmented Generation: an administrator uploads one or more reference PDF documents (such as this manual) through the chatbot's Context Manager panel. Each uploaded PDF is parsed page by page; the text of every page is split into overlapping chunks of roughly 800 characters (with a 200-character overlap between consecutive chunks) so that no single chunk loses too much surrounding context. These chunks are indexed using BM25, a keyword-relevance ranking algorithm. When a user submits a question, the system tokenizes the question, scores every indexed chunk against it, and retrieves the most relevant handful of chunks. Those chunks are then inserted into the prompt sent to the underlying language model — by default a locally hosted Ollama "llama3.2" model, or a configured cloud model — which composes a final answer grounded in that retrieved text and can cite the originating page numbers.

Because retrieval works page-by-page and chunk-by-chunk, a knowledge document is most effective for this assistant when each section restates enough context to stand on its own — which is why this manual consistently repeats the module and screen name at the start of each section rather than relying on the reader to remember it from a few pages earlier.

14.2 Using the Chat

Described fully in Part 3.3. In summary: the floating launcher in the bottom-right corner opens a chat panel with session history on the left, a message thread in the center, and a text input at the bottom. Suggested starter prompts are shown when a new session has no messages yet. The assistant maintains separate, independently deletable chat threads ("sessions") per user.

14.3 Context Manager (Administrators Only)

A settings panel within the chatbot widget, reachable only to users with the Admin role (or Django staff/superuser status), shows live system status — whether the Ollama server is connected, its endpoint URL, the configured model name, and which knowledge document is currently active (its title and how many indexed text chunks it produced). From this panel an administrator can upload a new PDF ("Index New Workflow PDF") to replace or add to the assistant's knowledge base; only .pdf files are accepted, and a successful upload shows "PDF ingested successfully. Ready for queries!" Non-admin users see the message "🔒 Administrator credentials required to upload and re-index PDF documents." in place of the upload form.

14.4 Agentic Actions the Assistant Can Perform

Beyond answering questions, the assistant can call a small set of backend tools to take real actions on the user's behalf, subject to the same permission rules as the rest of the application. When a tool call succeeds, the response may include one or more "ui_actions" that the frontend automatically executes — for example navigating the browser to a different screen, showing a toast notification, or telling an already-open list screen (Employees, Projects, or Customers) to silently refresh its data.

Tool	What It Does
navigate_to_page	Directs the user's browser to a specific screen — Dashboard, Employee Master, Customer Master, Detailer Master, Project Master, the Steel Budget worksheet, the Structural Tracking Schedule, the Production Priority Schedule, RFQ Data Entry, the Dollar Dashboard, the Erection Takeoff module, or the Field Moment Connections (FMC) dashboard — and shows a confirmation toast such as "Opened Employee Master".
list_projects	Retrieves a list of projects from the system so the assistant can answer questions like "what projects do we have" or "show me project X" directly in the chat.
get_employee_details	Looks up a specific employee's record (e.g. to answer "what is John's phone number").
create_employee	Creates a new Employee Master record directly from the conversation (e.g. "add a new employee named ..."), then refreshes the Employee Master screen if it is open.
create_customer	Creates a new Customer Master record — including an optional primary contact (name, email, phone) — directly from the conversation, then refreshes the Customer Master screen if it is open.

These actions are performed through the same backend APIs as the equivalent manual screens, so any validation rules described elsewhere in this manual for those screens apply equally when the assistant performs the action on a user's behalf.

PART 15 — END-TO-END BUSINESS WORKFLOWS

This Part stitches together the module-by-module descriptions in Parts 2–14 into the complete journeys a user actually follows in practice. Each journey below references the specific screen and Part where every step is documented in full detail.

15.1 From a New Enquiry to a Won Bid

17. A sales contact or estimator opens RFQ Master → Data Entry (Part 5.2) and clicks "+ New RFQ," entering the Quote No, Customer, Scope of Work, and Primary Estimator.
18. The team reviews the new bid's preparation timeline on the Internal Bid Schedule calendar (Part 5.9), which is automatically aware of company holidays from the Holiday Calendar (Part 5.10).
19. The assigned estimator opens Estimation & Erection (Part 6), selects the RFQ from the project selector, and works through Contracts (6.2), the Estimation Model worksheet (6.3), Erection Take-off (6.4), Field Moment Connections (6.5), and Misc Metals (6.6) to build every cost component.
20. The Breakdown tab (6.7) and the Estimate Data tab's 14 Cost Codes / Schedule of Values (6.8) give the estimator a final sanity check on the rolled-up numbers before submission.
21. Back on RFQ Master → Data Entry, the estimator fills in the remaining pricing/quantity columns directly on the spreadsheet and eventually sets Won/Lost to "Won" once the client accepts, optionally using "Set Job #" to assign the SFE Job Number.
22. If the bid needs to be re-quoted later (for example a value-engineering revision), "Duplicate as Rebid" (Part 5.2) clones the entire record into a fresh Quote Number rather than overwriting the original. Alternatively, the Estimation Summary screen's "Rebid Project" action (Part 5.11) reopens an existing estimate for revision.

15.2 From a Won Bid to an Active Project

23. In Project Master (Part 9), the user clicks "Add Project" and uses "Pre-fill Project from Won RFQs" to select the just-won RFQ, which auto-populates the project's name, code, customer, total tonnage, and total man-hours.
24. On the same modal's Structural Schedule tab (Part 9.3), the user builds out the project's sequence-by-sequence plan: sequence numbers, tonnage per sequence, item descriptions, and the full set of planned dates (Scheduled OFA/BFA/Field Measure, RTS, Scheduled Start of Erection).
25. Optionally, granular bill-of-material detail for any sequence is captured via the Fabrication Detail sub-modal (Part 9.4).
26. The identical sequence data is also visible — and can equally be created or edited — from Structural Schedule → Plan Creation (Part 10.1), which adds an inline Gantt-chart preview and a one-click "Structural Plan" PDF export per project.

15.3 Tracking Progress Through Detailing, Fabrication, and Erection

27. As shop drawings move through approval, the team records actual dates against each sequence on Structural Schedule → Plan Tracking (Part 10.2) — for example Actual RTS Date — which the system validates cannot be set in the future.
28. The Dashboard's Operations view (Part 4.2) automatically reclassifies every sequence into Under Detailing, Fabrication, or Sent for Erection based on which actual dates have been recorded, and flags any sequence whose planned RTS date has passed as Likely Delay or Delayed in the Project Health Hierarchy panel.

29. Once a sequence is ready to be physically built, it is grouped into a named batch on Production Management → Production Priority Schedule (Part 11.2), with a defined start/end window for that batch.
30. Within that batch, Process Master (Part 11.3) routes each item through its specific machine stations across the Plate, Angle, or Structural priority categories, recording Actual RTS Date, Run Days, and Start/Complete Run dates per item.
31. Process Master Settings (Part 11.4) governs the dollar rate associated with each of the nine routing stations used in that routing step.

15.4 Balancing Shop Capacity Against the Pipeline

32. Capacity Mapping (Part 12) is kept current independently of any single project: Machine Master (12.3) registers the shop's equipment, Workforce Master (12.4) registers available labor by skill level and month, and Capacity Configuration (12.2) defines the resulting raw tonnage/hour capacity per shop and process.
33. The shared Performance Overview (12.1) then automatically compares that available capacity against the man-hour demand generated by every currently Won RFQ's fabrication/erection hours for the selected month(s), surfacing a Surplus or Shortage figure leadership can act on.
34. The same forward-looking comparison, extended to a rolling 18 months and broken out by trade, is available on RFQ Master → Future Capacity (Part 5.7), which also distinguishes "Potential" demand (all open, non-Lost bids) from "Confirmed" demand (Won bids only).

15.5 Finalizing and Locking an Estimate

Once an estimate is considered final, the team uses the Estimate Data tab's "Finalize Estimate" workflow (Part 6.8): both Prevailing Wage and Out-of-Town flags are confirmed, and an authorized preparer types their full name as a digital signature into the Finalize Estimate Submission modal. Confirming creates an immutable, timestamped revision snapshot (visible afterward in the Revision History panel, downloadable as PDF or XLSX) and switches every tab across the entire Estimation & Erection module for that project to read-only. Any further changes require cloning the estimate into a new revision rather than editing the locked one in place.

15.6 Executive Oversight

Throughout all of the above, users with the Manager or Admin role (and, in a single-purpose way, Estimator, Detailing, and Read-Only roles) can step back from any individual screen and view the aggregate picture: the Dashboard's Executive/VP view (Part 4.3) for bid pipeline, win-rate, and forward capacity outlook; RFQ Master's five analytical tabs (Bid Performance, Dollar Dashboard, Job Analytics, Sales Cycle, and Future Capacity, Parts 5.3–5.7) for deeper financial and historical-benchmark analysis; and the Print Setup screen (Part 5.8) for a clean, printable weekly bid-review hand-out.

PART 16 — STATUS, VALIDATION & KNOWN-LIMITATIONS REFERENCE

This Part consolidates, in one place, the formal status values used by each data type in the system, plus a summary of validation patterns that recur across many screens, and an honest account of a couple of screens that exist in the application's code but are not currently reachable through normal navigation — included here so the assistant never claims a user can do something that the live application does not actually expose.

16.1 Status Values By Data Type

Data Type	Allowed Status / Choice Values
Project (Project Master / Plan Creation)	Planning · In Progress · Yet to Start · Completed (a project is only treated as "finalized," locking its Estimation & Erection module, once status = Completed)
Project Priority	Low · Medium · High
Structural Sequence schedule-adherence (Dashboard, Production Gantt)	Yet to Start · On Track · Completed · Completed with Delay · Likely Delay (within 7 days past plan) · Delayed (more than 7 days past plan)
Employee Status	Active · On Leave · Inactive
RFQ / Bid Won/Lost	Pending (displayed as "In Progress" in some summary counters) · Won · Lost (Cancelled also appears as a status on the Internal Bid Schedule)
RFQ Decision to Bid	Yes · No · TBD (default)
RFQ Scope of Work	Fabrication · Detailing · Erection
Holiday Type	Public · Company · Optional
Production Priority Module Type	PLATE · ANGLE · STRUCTURAL
Manpower Skill Level	High · Medium · Low
Capacity Category	Machine · Manual
User Role	admin · manager · employee · estimator · detailing · readonly (see Part 1.5.1)
Milestone (data model only — see 16.3)	Pending · In Progress · Completed · Overdue, with Priority Low · Medium · High · Critical

16.2 Common Validation Patterns

- **Required-field guards:** most create forms refuse to submit if their core identifying fields are blank — for example Project Name and Code on Project Master/Plan Creation ("Project Name and Code are required"), Schedule Number/Start Date/End Date on a Production Schedule, or Title/Message/From Date/To Date on an Announcement.
- **Password rules:** every place a password is set (Reset Password, Settings → Security, User Access registration/edit) enforces a minimum of 6 characters and requires the confirmation field to match exactly.
- **Email format:** User Access validates email addresses against a basic pattern and reports "Invalid email address" if it does not look like an email.
- **Phone format:** the Employee Master phone field strips non-digit characters as the user types and enforces exactly 10 digits.

- **Date-in-the-past guards:** Plan Tracking refuses to let a user record any "actual" milestone date in the future ("Actual dates cannot be set in the future.").
- **Delete confirmations:** every delete action in the system requires an explicit confirmation step — either a native browser confirm dialog or an in-app confirmation modal — before the record is removed; there is no "undo" after confirming.

16.3 Known Limitations — Screens Present But Not Currently Linked

In the interest of accuracy, two screens exist in the application's source code but are not reachable through the normal Sidebar navigation in the current build, and should not be described to a user as available features unless this changes in a future release:

- **"Milestone Management" page:** a stand-alone Milestone Management screen exists in the codebase (with status filter tabs, a search box, and an "Add Milestone" modal) but is not wired into the application's routes or Sidebar, and its data is currently a static example data set rather than live records — it cannot actually be opened by a user today. The application's real, functional equivalent of project milestone visualization is the "Milestone Management" Gantt-chart widget on the Operations Dashboard (Part 4.2.3), and day-to-day milestone/date tracking is otherwise done through the Structural Schedule grid (Part 9.3) and Plan Tracking (Part 10.2).
- **Legacy "Bid Enquiry" page:** a second, older Bid Enquiry screen (distinct from RFQ Master's Data Entry tab) exists at the URL `"/bids/enquiry"` and remains technically reachable by a user who has RFQ-module access and types that address directly, but it has no entry in the Sidebar and is not part of the documented day-to-day workflow. The current, supported way to log and manage RFQs is the RFQ Master → Data Entry tab described in Part 5.2.

16.4 Notification Bell

The notification Bell in the Header (Part 3.2) is present on every screen, but as of this version of the application no notification-generating events have been wired up yet — the dropdown will always read "No new notifications — You're all caught up!" This is expected behavior, not an error.

PART 17 — APPENDICES

Appendix A — Complete Sidebar-to-Route Map

Every navigation item in the Sidebar (Part 3.1), its underlying URL, and the Part of this manual where it is documented.

Sidebar Section	Item	Route	Documented In
OVERVIEW	Dashboard	/dashboard	Part 4
BID MANAGEMENT	RFQ Master	/rfq (→ /rfq/data-entry)	Part 5.1–5.8
BID MANAGEMENT	Internal Bid Schedule	/bids/schedule	Part 5.9
BID MANAGEMENT	Holiday Calendar	/bids/holidays	Part 5.10
BID MANAGEMENT	Estimation Summary	/estimation-summary	Part 5.11
BID MANAGEMENT	Estimation	/estimation-erection (→ estimation)	Part 6
MANAGEMENT	Employee Master	/employees	Part 8.1
MANAGEMENT	Customer Master	/customers	Part 8.2
MANAGEMENT	Detailer Master	/detailers	Part 8.3
BUDGET ESTIMATOR	Design Inputs	/steel-budget/input	Part 7.1
BUDGET ESTIMATOR	Estimation Result	/steel-budget/result	Part 7.2
PROJECT MANAGEMENT	Project Master	/projects	Part 9
STRUCTURAL SCHEDULE	Plan Creation	/structural/plan-creation	Part 10.1
STRUCTURAL SCHEDULE	Plan Tracking	/structural/plan-tracking	Part 10.2
PRODUCTION MANAGEMENT	Production Schedule	/production (→ priority-schedule)	Part 11.1–11.3
PRODUCTION MANAGEMENT	Process Master Settings	/production/master-settings	Part 11.4
CAPACITY MAPPING	Capacity Configuration	/production/capacity-mapping/capacity	Part 12.2
CAPACITY MAPPING	Machine Master	/production/capacity-mapping/machine	Part 12.3
CAPACITY MAPPING	Workforce Master	/production/capacity-mapping/manpower	Part 12.4
SYSTEM	Settings	/settings	Part 13.1
SYSTEM	Announcement	/announcements	Part 13.2
SYSTEM	User Access (Admins only)	/user-access	Part 13.3

Two additional always-public routes sit outside this map: `"/login"` and `"/forgot-password" / "/reset-password"` (Part 2). All routes are mounted under the application base path `"/SFE"`.

Appendix B — Complete Module-Key Permission List

These are the exact module-key strings used by the per-user `"allowed_modules"` permission list described in Part 1.5.2 and managed on the User Access screen (Part 13.3). A new user account is automatically granted every key in this list except `"User Access"` itself, which is never granted to non-Admins regardless of this list.

- `/dashboard`
- `/rfq`
- `/bids/schedule`
- `/bids/holidays`
- `/estimation-summary`
- `/estimation-erection`
- `employees`
- `customers`
- `detailers`
- `steel-budget/input`
- `steel-budget/result`
- `projects`
- `/structural/plan-creation`
- `/structural/plan-tracking`
- `/production/priority-schedule`
- `/production/process-master`
- `/production/capacity-mapping/capacity`
- `/production/capacity-mapping/machine`
- `/production/capacity-mapping/manpower`
- `settings`
- `announcements`

Appendix C — Glossary of Terms

Term	Meaning
RFQ	Request for Quotation — a prospective job a customer has asked SFE to bid on.
SOV	Schedule of Values — the standard line-item cost breakdown (Bond, Drawings, Steel, Joist, Deck, Misc Metals, Erection) used for billing/draw purposes.
OFA	Originals-For-Approval — the first issue of shop/erection drawings sent to the customer/engineer for review.
BFA	Back-For-Approval — the returned, approved (or commented) drawings after customer/engineer review.
RTS	Ready-To-Ship — the date a structural sequence is planned (or actually) ready to leave the shop for the job site.

Term	Meaning
AISC	American Institute of Steel Construction — source of the standard structural steel shape designations (e.g. W14X82) used throughout the Estimation & Erection module.
FMC	Field Moment Connection(s) — the Estimation & Erection tab covering bolted/welded moment-connection take-off.
SEBW	Steel Estimate / Bid Worksheet — an external pre-bid weight/cost worksheet whose figures can be pulled directly into an RFQ via the "Pull from SEBW" modal.
PE Stamp	Professional Engineer certification/stamping cost line item on the Estimation Model worksheet.
CCIP / OCIP	Contractor-Controlled / Owner-Controlled Insurance Program — a project-flag captured on the RFQ Data Entry grid.
Tonnage / Tons	The weight of structural steel, used as the primary unit for pricing, scheduling, and capacity throughout the system.
Manhour / Ton (MH/Ton)	A project-level efficiency ratio (Total Manhours ÷ Total Ton) auto-computed on Project Master and used to budget each sequence's shop hours.
Sequence (Seq #)	One row of a project's Structural Schedule — a discrete portion of the job's steel with its own tonnage, drawing, and date milestones.
RAG	Retrieval-Augmented Generation — the architecture behind the in-app AI assistant, described fully in Part 14.1.